

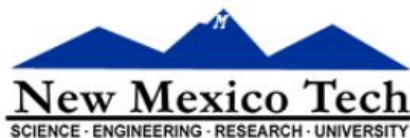
# Imaging and Deconvolution

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Fourteenth Synthesis Imaging Workshop

2014 May 13– 20



# References

- Thompson, A.R., Moran, J.M., Swensen, G.W. 2004  
“Interferometry and Synthesis in Radio Astronomy”, 2<sup>nd</sup> edition (Wiley-VCH)
- previous Synthesis Imaging Workshop proceedings
  - Perley, R.A., Schwab, F.R., Bridle, A.H. eds. 1989 ASP Conf. Series 6  
“Synthesis Imaging in Radio Astronomy” (San Francisco: ASP)
    - Ch. 6 Imaging (Sramek & Schwab) and Ch. 8 Deconvolution (Cornwell)
  - [www.aoc.nrao.edu/events/synthesis](http://www.aoc.nrao.edu/events/synthesis)
    - lectures by Cornwell 2002 and Bhatnagar 2004, 2006
- IRAM Interferometry School proceedings
  - [www.iram.fr/IRAMFR/IS/IS2008/archive.html](http://www.iram.fr/IRAMFR/IS/IS2008/archive.html)
    - Ch. 13 Imaging Principles and Ch. 16 Imaging in Practice (Guilloteau)
    - lectures by Pety 2004-2012
- many other lectures and pedagogical presentations are available
  - ALMA primer, ATNF, CARMA, ASIAA, e-MERLIN, ...



# Visibility and Sky Brightness

- $V(u,v)$ , the complex visibility function, is the 2D Fourier transform of  $T(l,m)$ , the sky brightness distribution (for incoherent source, small field of view, far field, etc.)  
[for derivation from van Cittert-Zernike theorem, see TMS Ch. 14]

- mathematically

$$V(u, v) = \int \int T(l, m) e^{-i2\pi(ul+vm)} dl dm$$

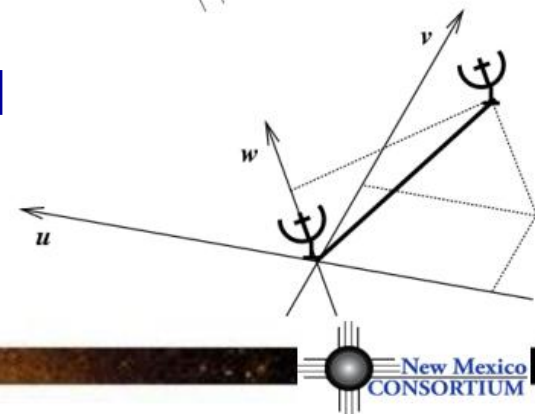
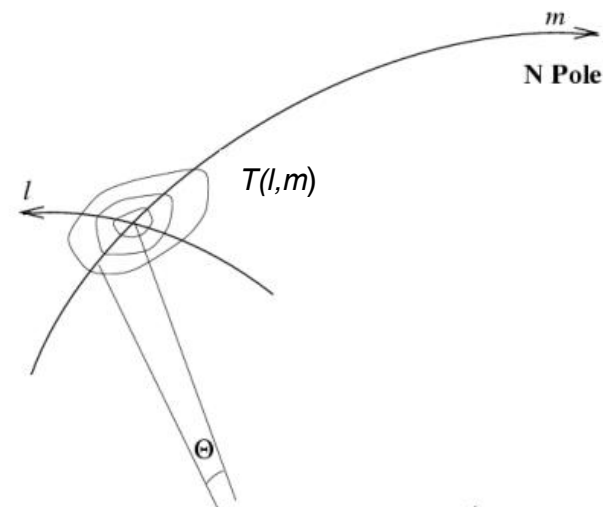
$$T(l, m) = \int \int V(u, v) e^{i2\pi(ul+vm)} du dv$$

$u, v$  are E-W, N-S spatial frequencies [wavelengths]

$l, m$  are E-W, N-S angles in the tangent plane [radians]

(recall  $e^{ix} = \cos x + i \sin x$ )

$$V(u, v) \xrightarrow{\mathcal{F}} T(l, m)$$

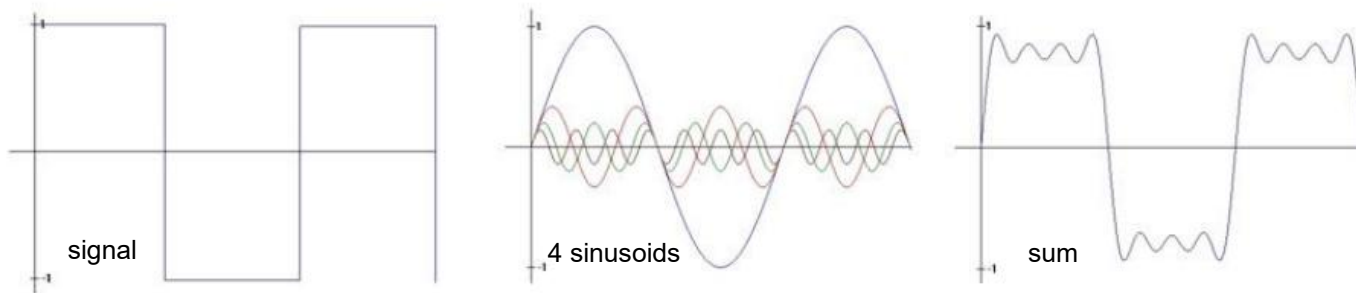


# The Fourier Transform

- Fourier theory states and any well behaved signal (including images) can be expressed as the sum of sinusoids



**Jean Baptiste  
Joseph Fourier**  
1768-1830



$$x(t) = \frac{4}{\pi} \left( \sin(2\pi ft) + \frac{1}{3} \sin(6\pi ft) + \frac{1}{5} \sin(10\pi ft) + \dots \right)$$

- the Fourier transform is the mathematical tool that decomposes a signal into its sinusoidal components
- the Fourier transform contains *all* of the information of the original signal

# The Fourier Domain

- acquire some comfort with the Fourier domain
- in older texts, functions and their Fourier transforms occupy *upper* and *lower* domains, as if “functions circulated at ground level and their transforms in the underworld” (Bracewell 1965)



- some properties of the Fourier transform  $g(x) \xrightarrow{\mathcal{F}} G(s)$

adding  $g(x) + h(x) = G(s) + H(s)$

scaling  $g(\alpha x) = \alpha^{-1} G(s/\alpha)$

shifting  $g(x - x_0) = G(s) e^{i2\pi x_0 s}$

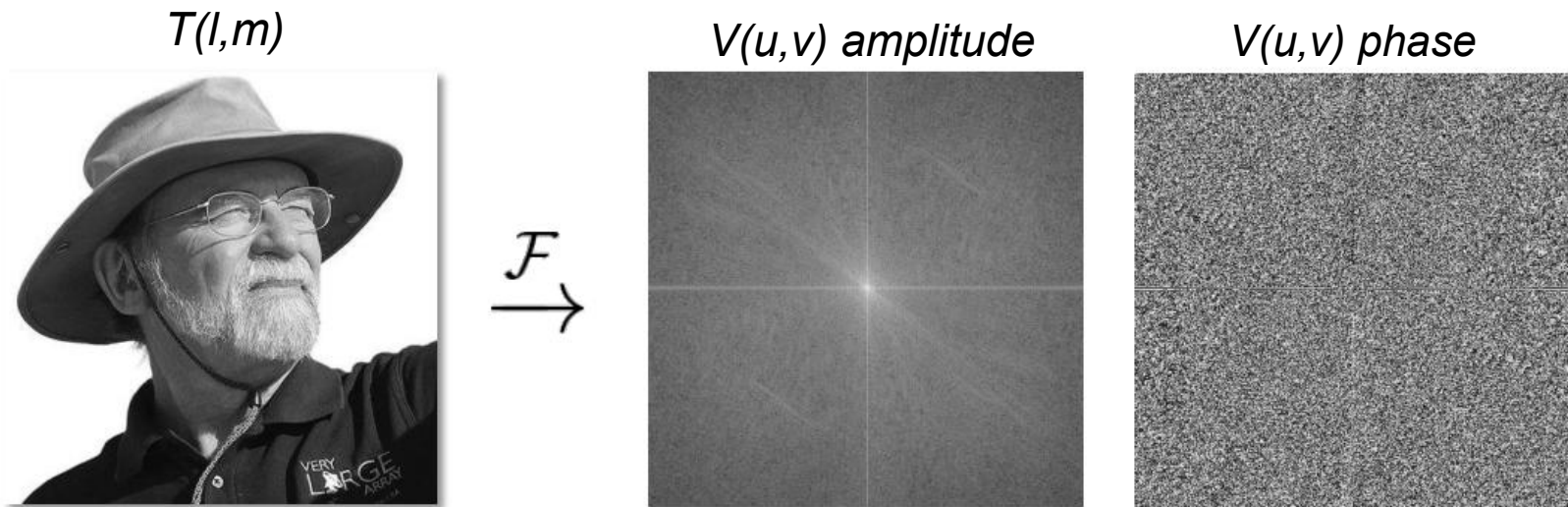
convolution/multiplication  $g(x) = h(x) * k(x) \quad G(s) = H(s)K(s)$

Nyquist-Shannon sampling theorem

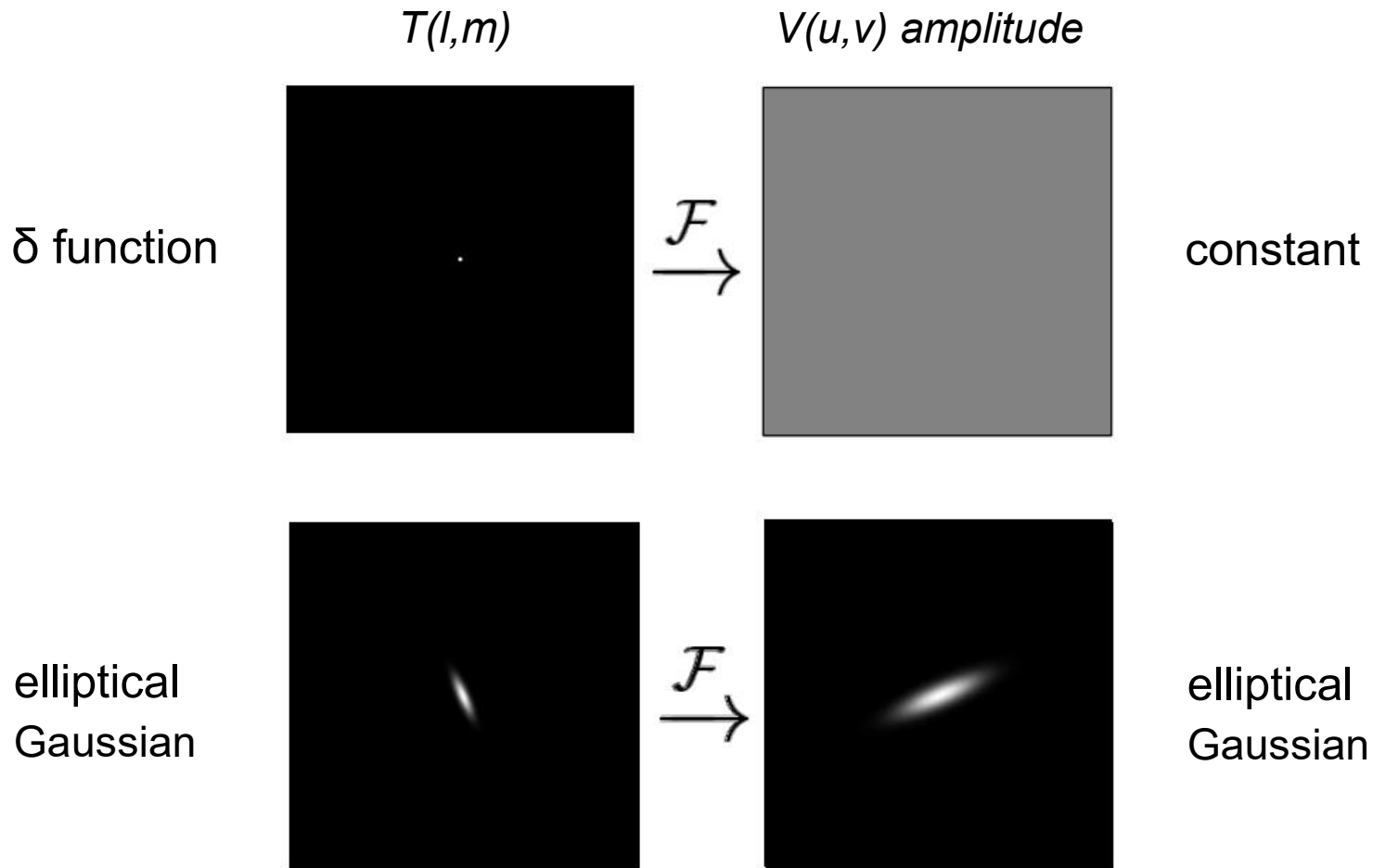
$g(x) \subset \Theta$  completely determined if  $G(s)$  sampled at  $\leq 1/\Theta$

# Visibilities

- each  $V(u,v)$  contains information on  $T(l,m)$  *everywhere*, not just at a given  $(l,m)$  coordinate or within a particular subregion
- each  $V(u,v)$  is a complex quantity
  - expressed as (*real, imaginary*) or (*amplitude, phase*)



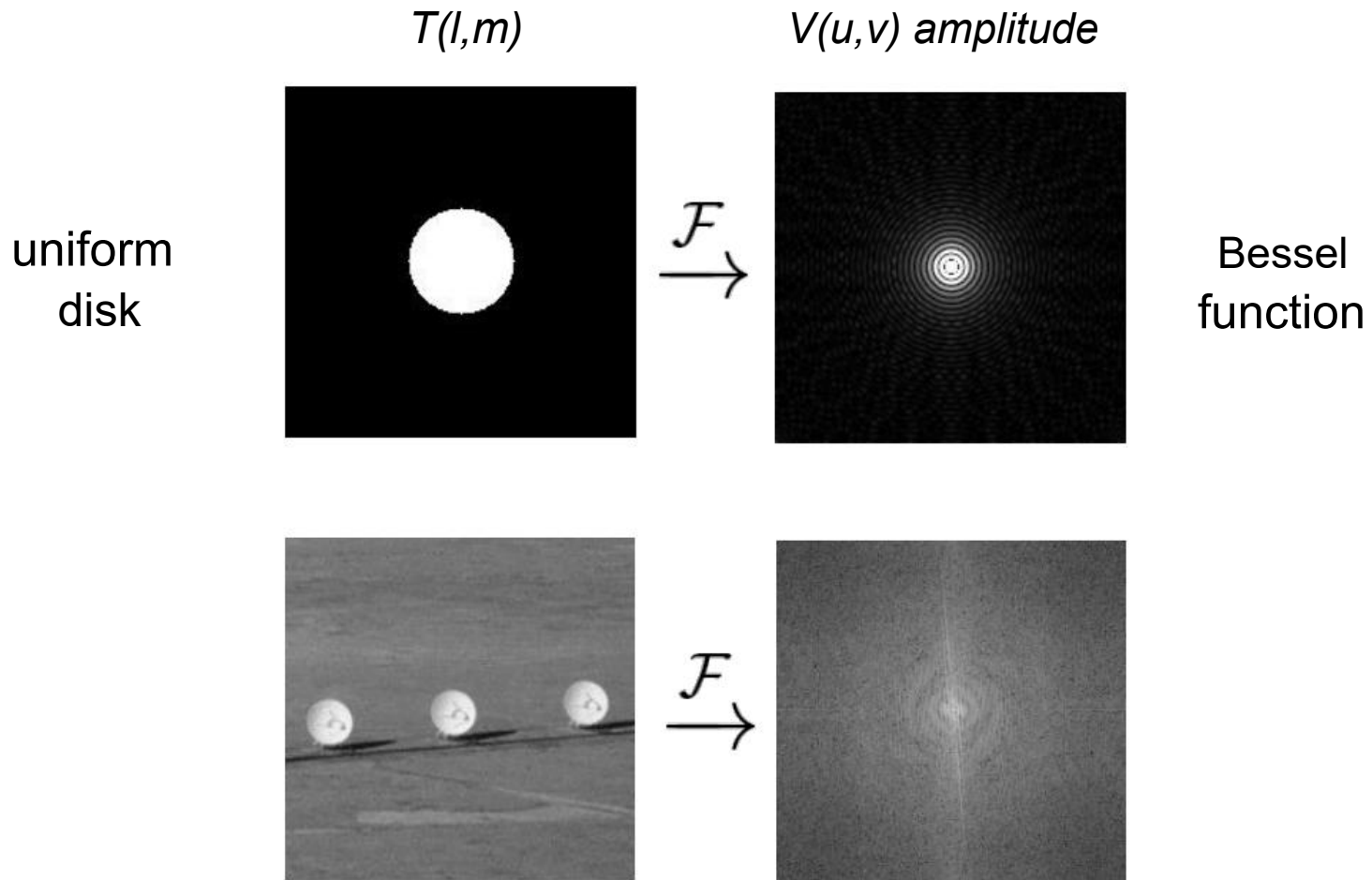
# Example 2D Fourier Transforms



*narrow features transform into wide features (and vice-versa)*



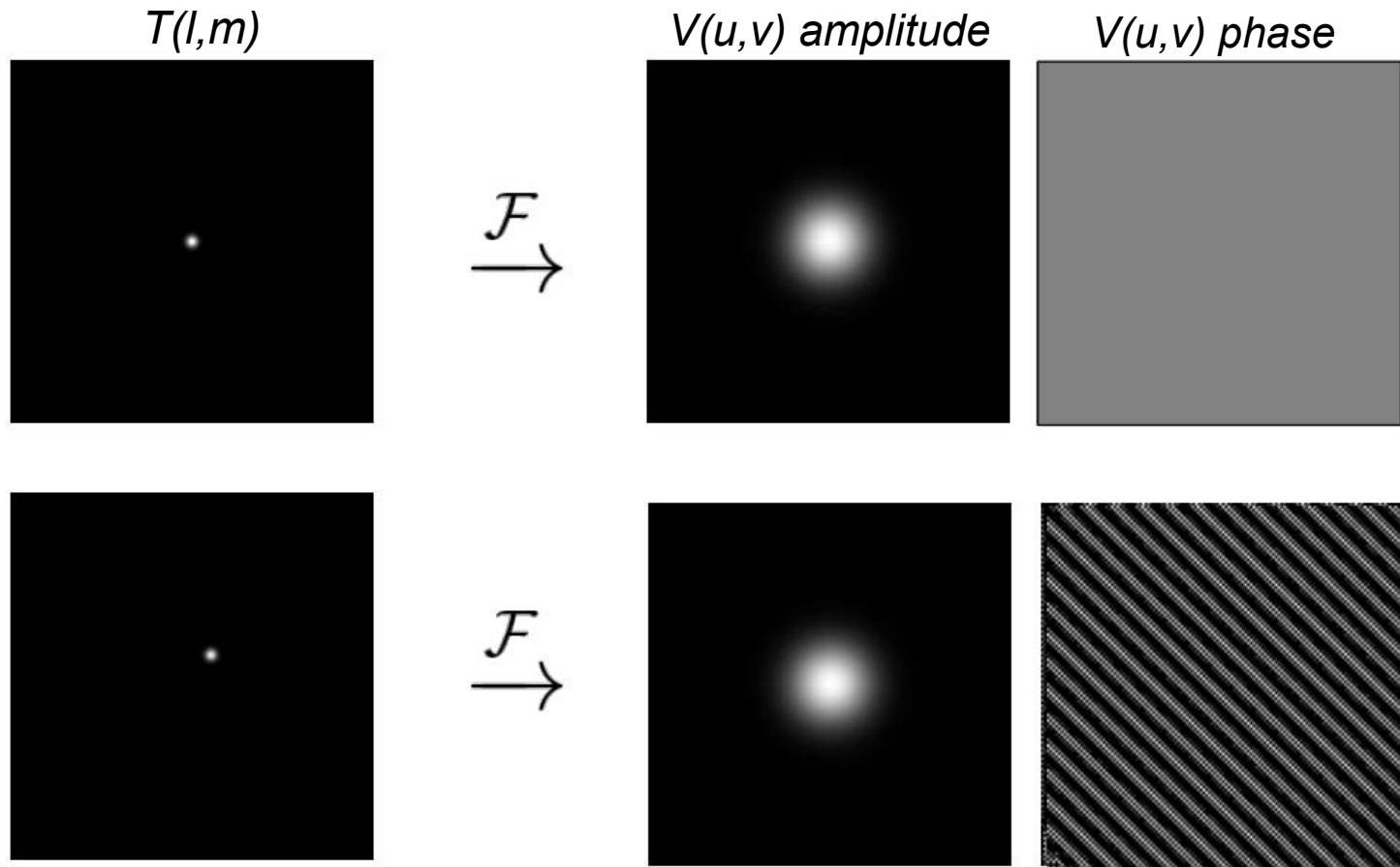
# Example 2D Fourier Transforms



*sharp edges result in many high spatial frequencies*

# Amplitude and Phase

- amplitude tells “how much” of a certain spatial frequency
- phase tells “where” this spatial frequency component is located



# The Visibility Concept

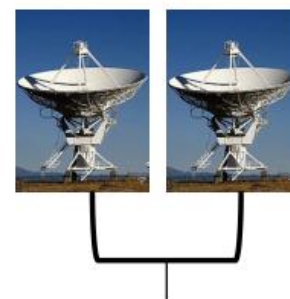
$$V(u, v) = \int \int T(l, m) e^{-i2\pi(ul+vm)} dl dm$$

- visibility as a function of baseline coordinates  $(u, v)$  is the Fourier transform of the sky brightness distribution as a function of the sky coordinates  $(l, m)$
- $V(u=0, v=0)$  is the integral of  $T(l, m) dl dm =$  total flux density
- since  $T(l, m)$  is real,  $V(-u, -v) = V^*(u, v)$ 
  - $V(u, v)$  is Hermitian
  - get two visibilities for one measurement



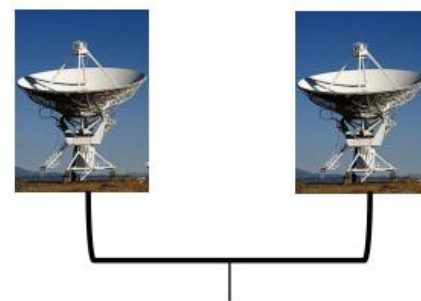
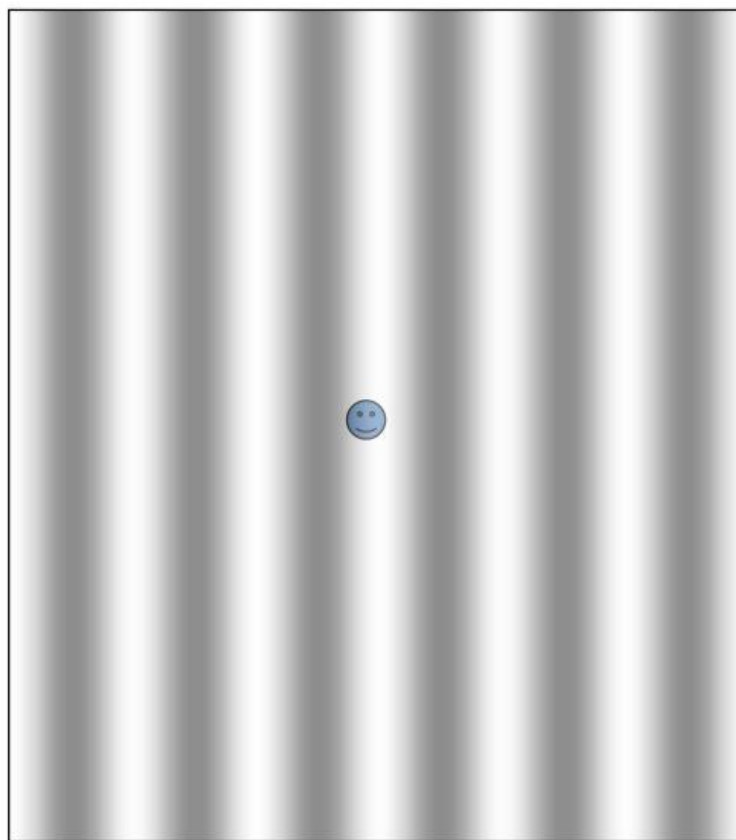
# The Visibility Concept

$$V(u, v) = \iint T(l, m) e^{-i2\pi(ul+vm)} dl dm$$



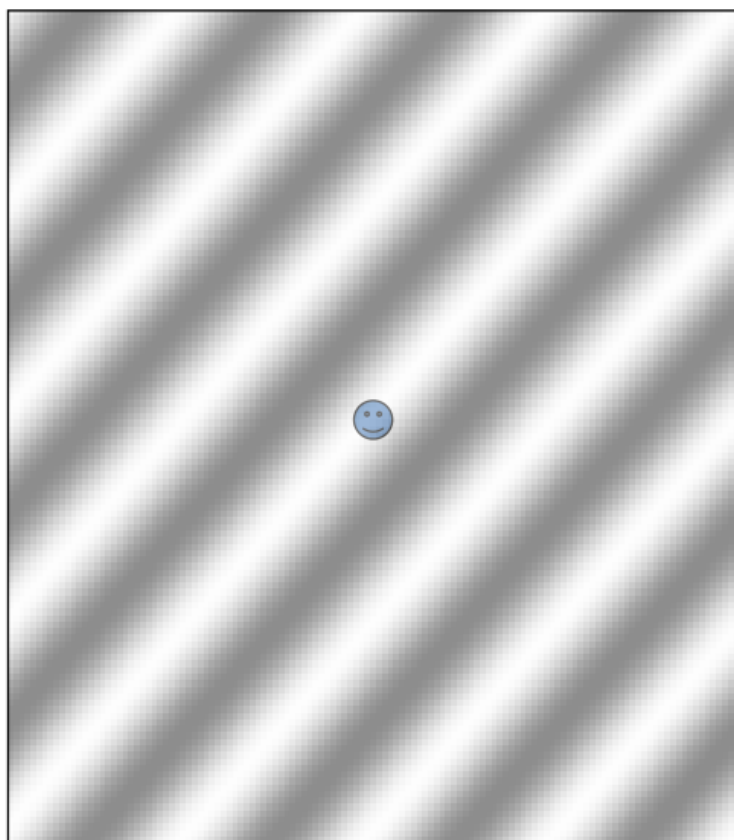
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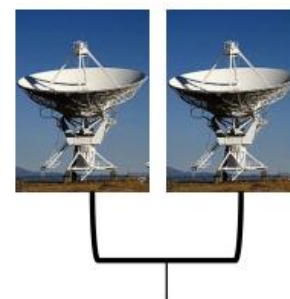
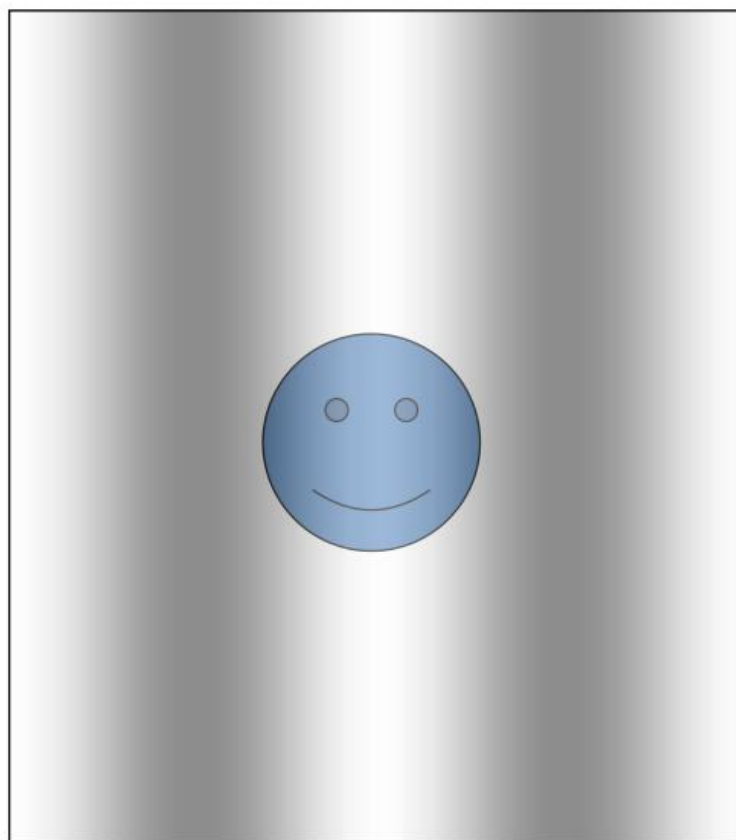
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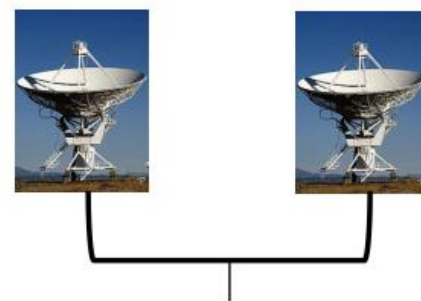
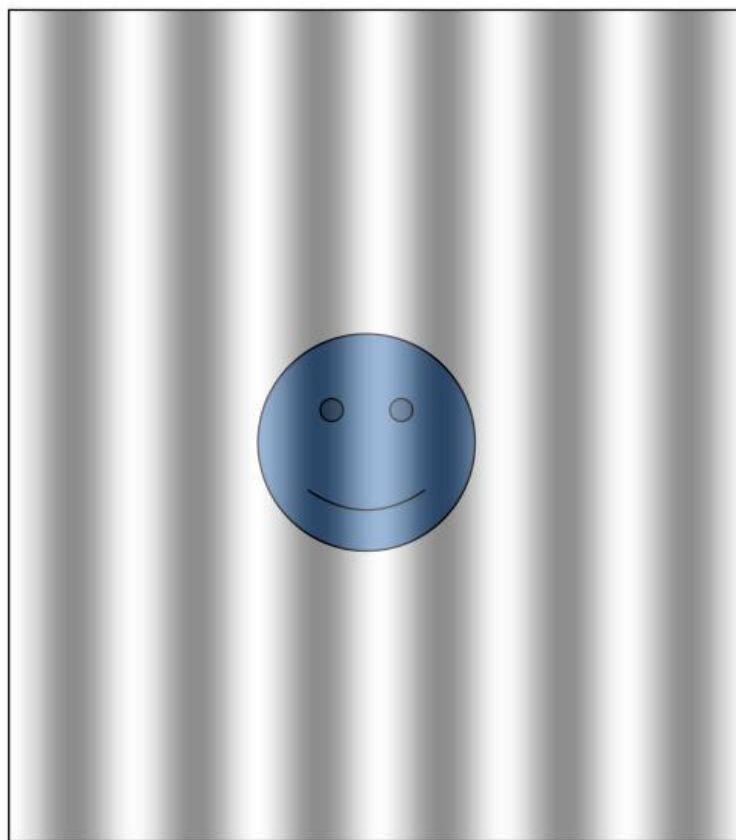
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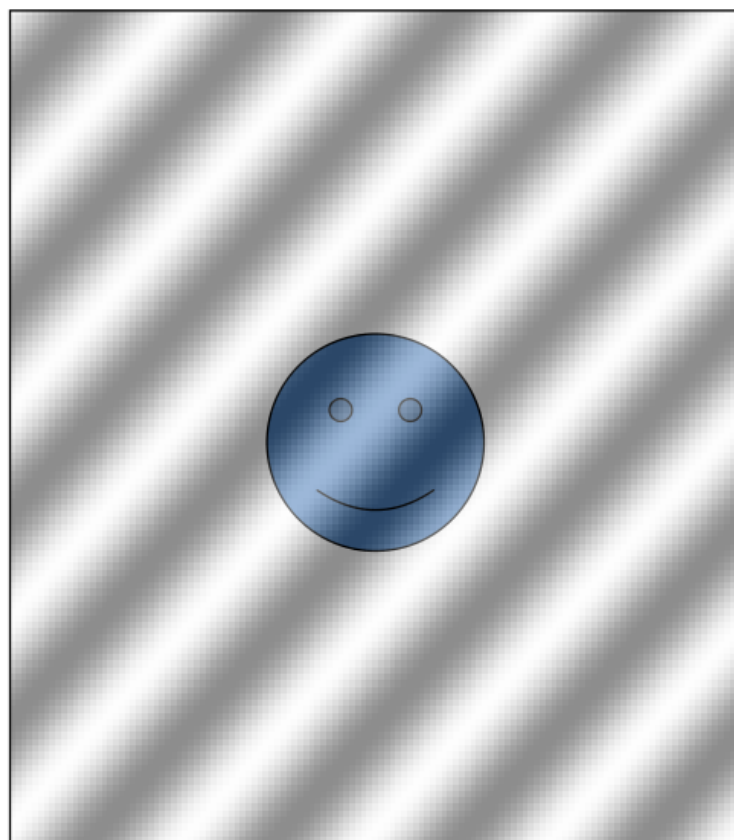
# The Visibility Concept

$$V(u, v) = \iint T(l, m) e^{-i2\pi(ul+vm)} dl dm$$



# The Visibility Concept

$$V(u, v) = \iint T(l, m) e^{-i2\pi(ul+vm)} dl dm$$



# Aperture Synthesis Basics

- idea: sample  $V(u,v)$  at enough  $(u,v)$  points using distributed small aperture antennas to synthesize a large aperture antenna of size  $(u_{max}, v_{max})$
- one pair of antennas = one baseline  
= two  $(u,v)$  samples at a time
- $N$  antennas =  $N(N-1)$  samples at a time
- use Earth rotation to fill in  $(u,v)$  plane over time  
(Sir Martin Ryle, 1974 Nobel Prize in Physics)
- reconfigure physical layout of  $N$  antennas for more samples
- observe at multiple wavelengths for  $(u,v)$  plane coverage, for source spectra amenable to simple characterization (“multi-frequency synthesis”)
- if source is variable, then be careful



**Sir Martin Ryle**  
1918-1984

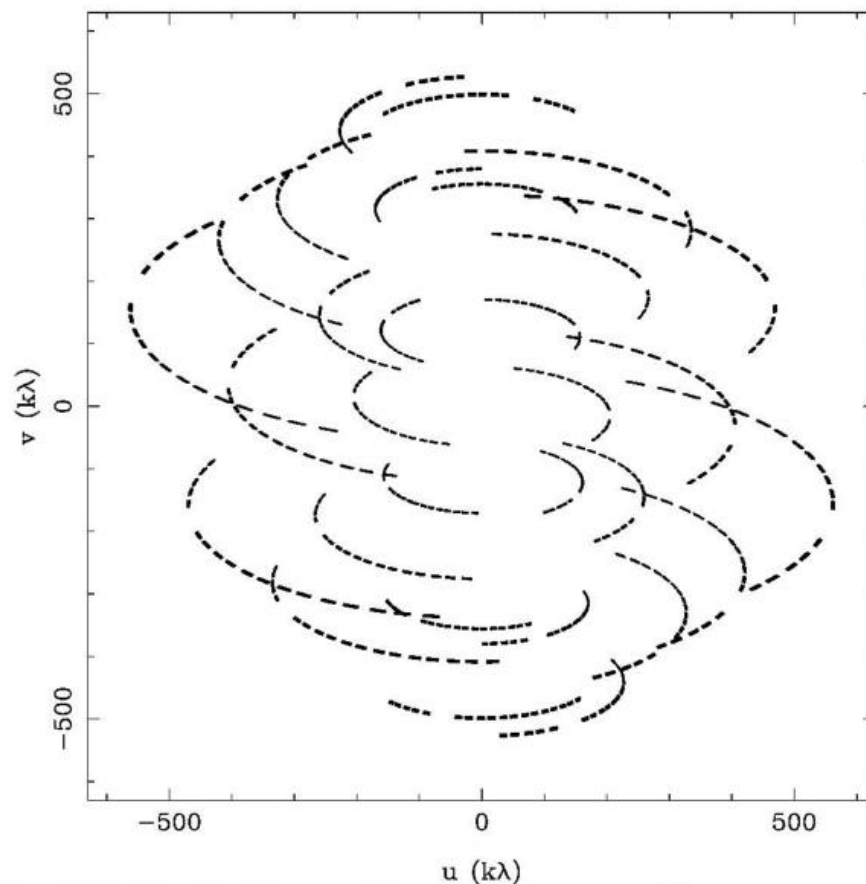
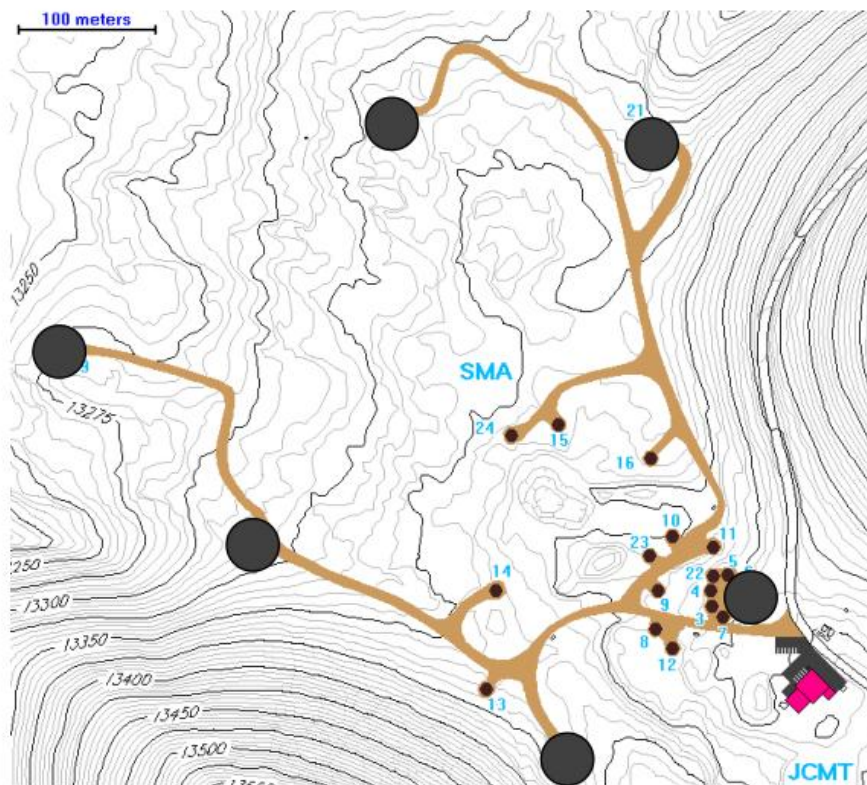


# Examples of Aperture Synthesis Telescopes (for Millimeter Wavelengths)



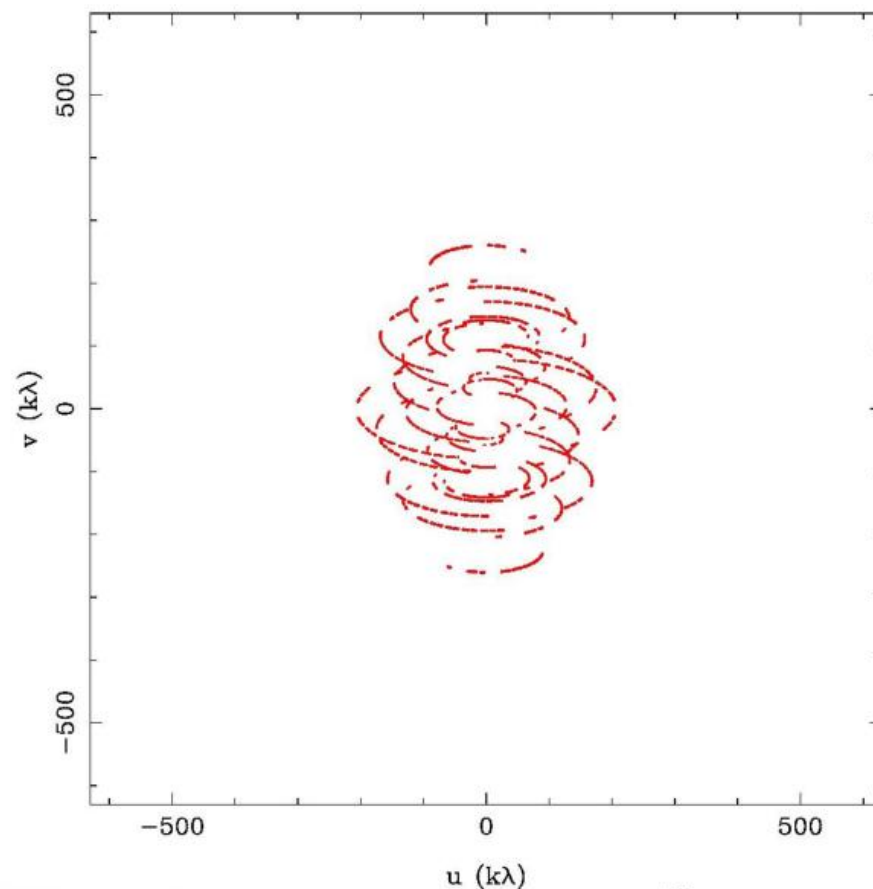
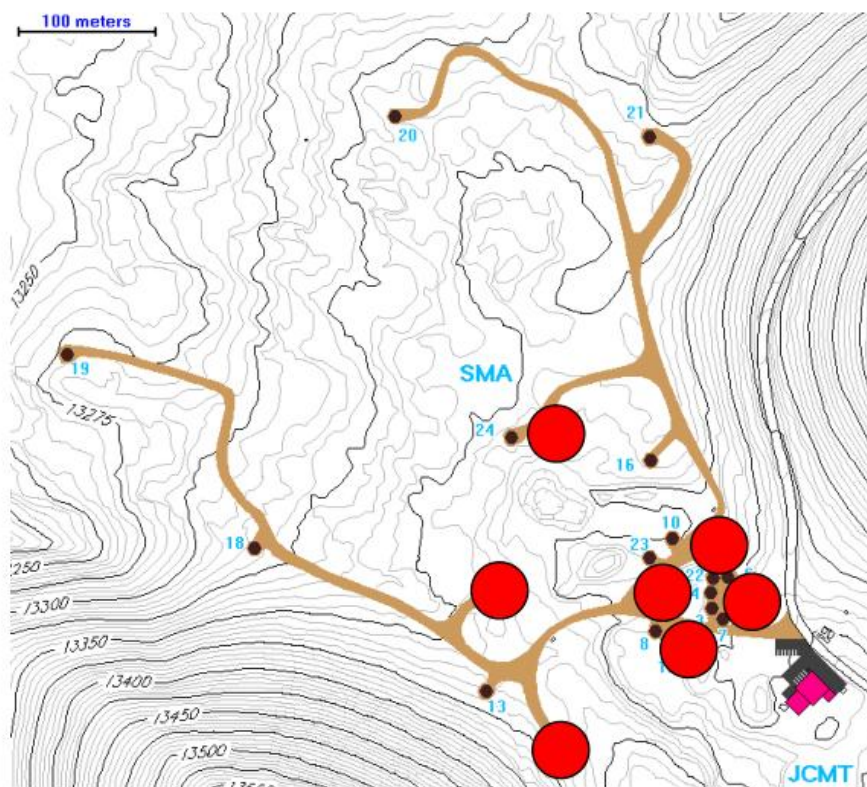
# An Example of (u,v) plane Sampling

VEX configuration of 6 SMA antennas,  $\nu = 345$  GHz, dec = +22 deg



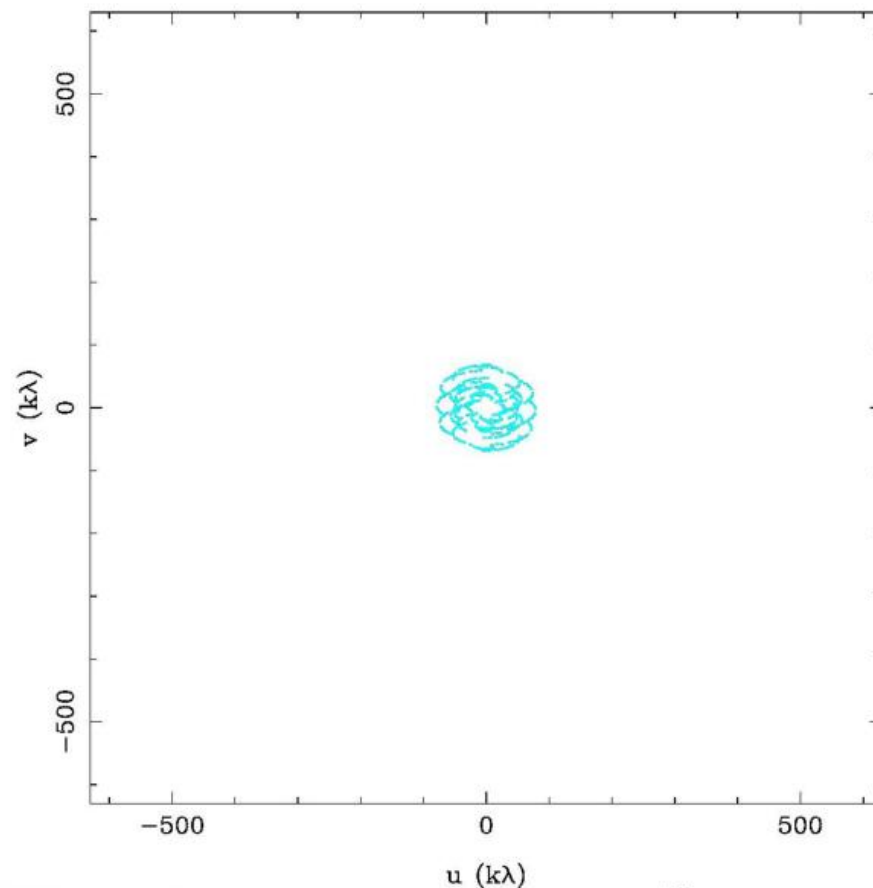
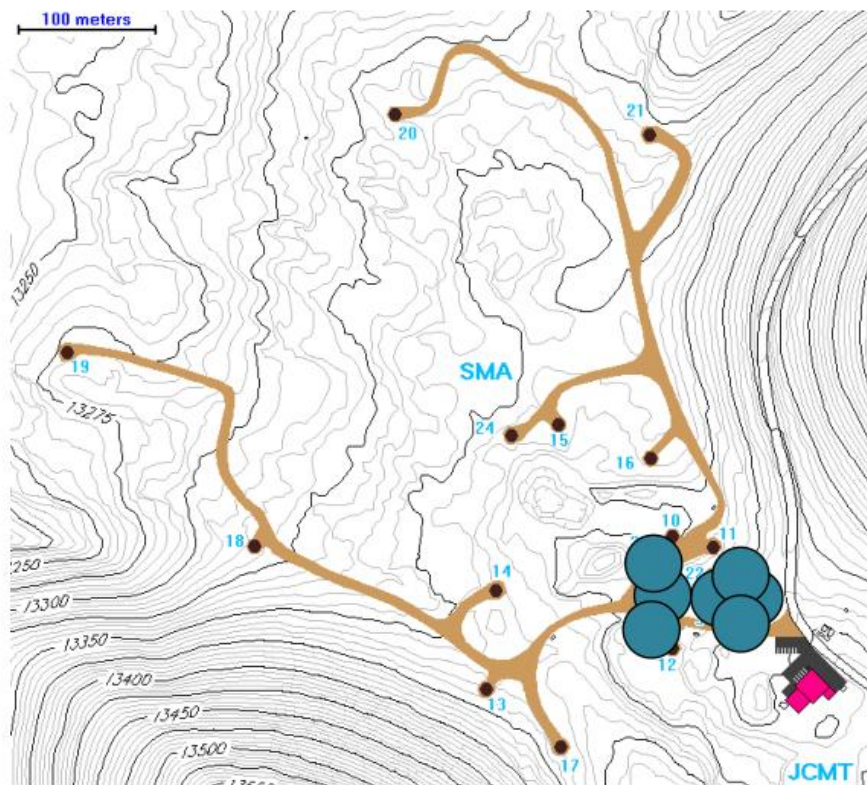
# An Example of (u,v) plane Sampling

EXT configurations of 7 SMA antennas,  $\nu = 345$  GHz,  $\text{dec} = +22^\circ$



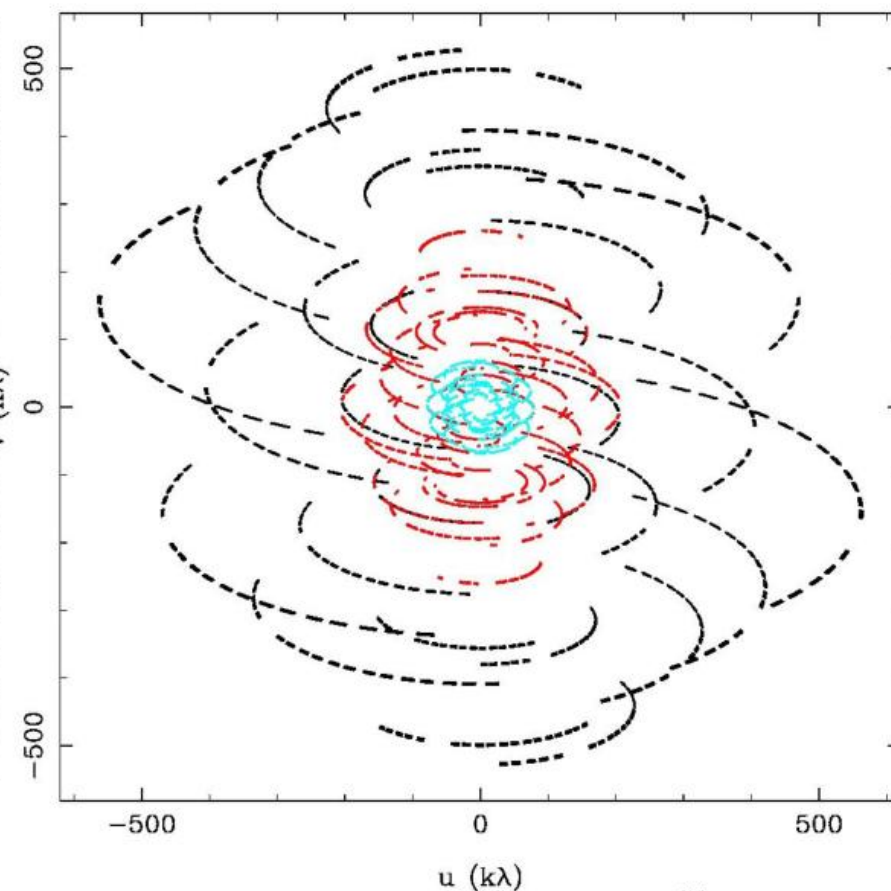
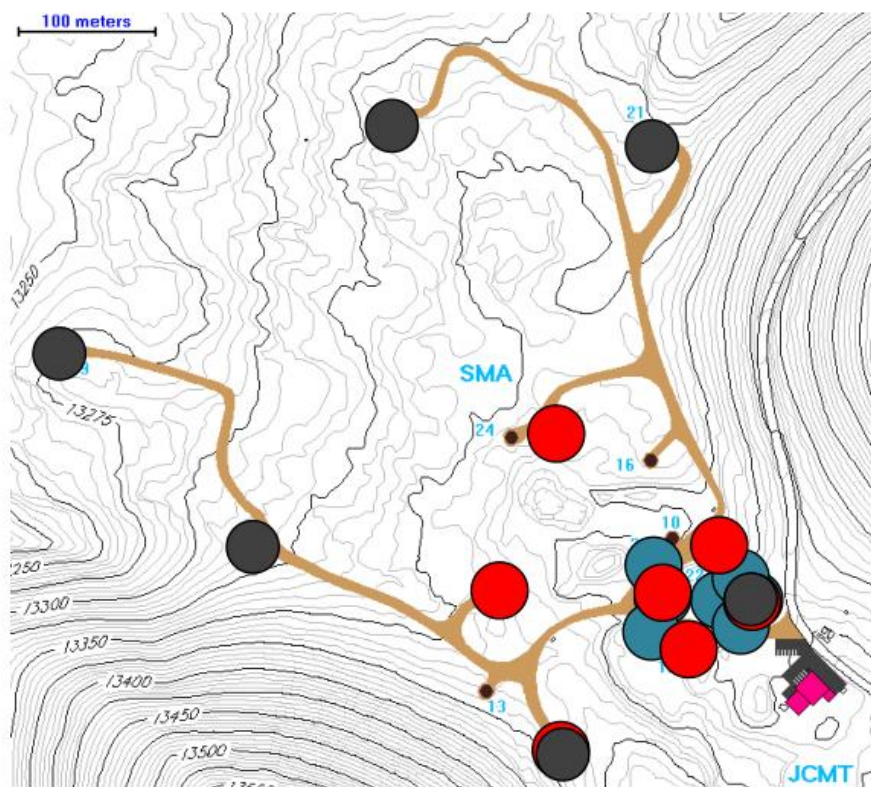
# An Example of (u,v) plane Sampling

COM configurations of 7 SMA antennas,  $\nu = 345$  GHz,  $\text{dec} = +22^\circ$



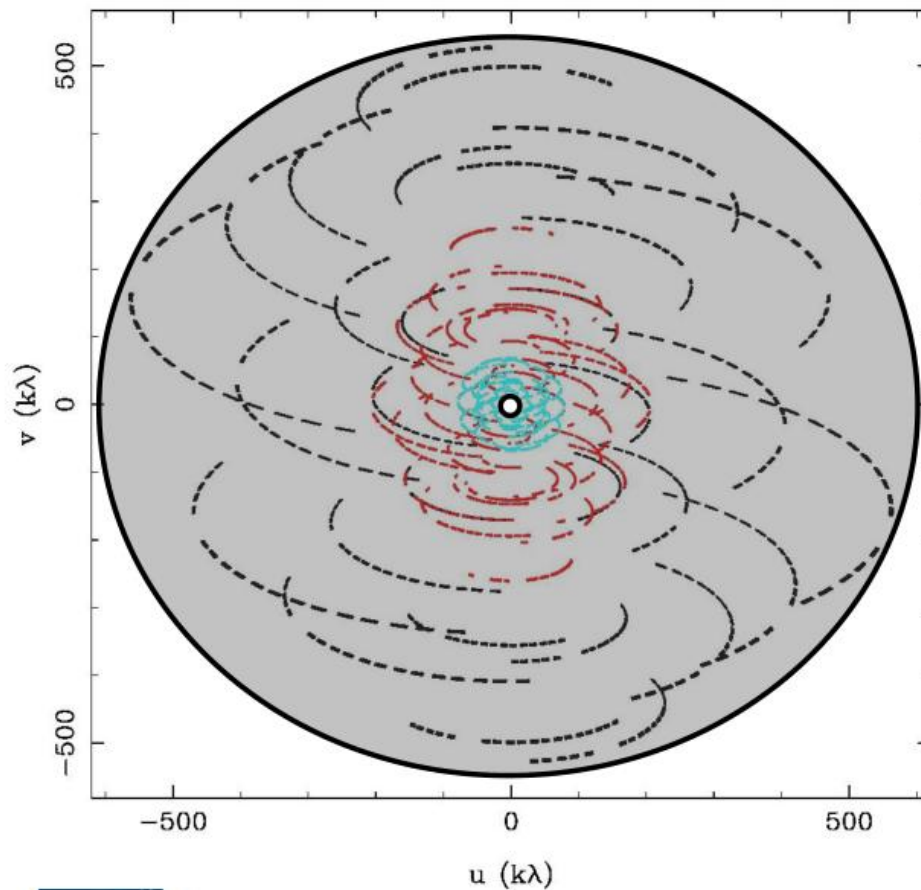
# An Example of (u,v) plane Sampling

3 configurations of SMA antennas,  $\nu = 345$  GHz,  $\text{dec} = +22^\circ$



# Implications of (u,v) plane Sampling

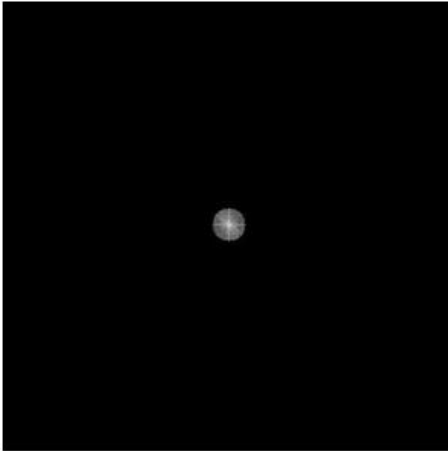
samples of  $V(u,v)$  are limited by number of antennas and by Earth-sky geometry



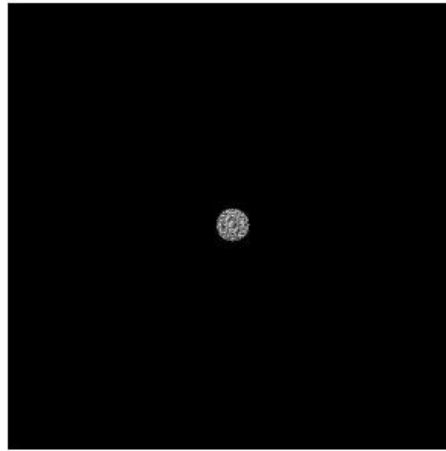
- *outer boundary*
  - no information on smaller scales
  - resolution limit
- *inner hole*
  - no information on larger scales
  - extended structures invisible
- *irregular coverage between boundaries*
  - sampling theorem violated
  - information missing

# Inner and Outer (u,v) Boundaries

$V(u,v)$  amplitude



$V(u,v)$  phase

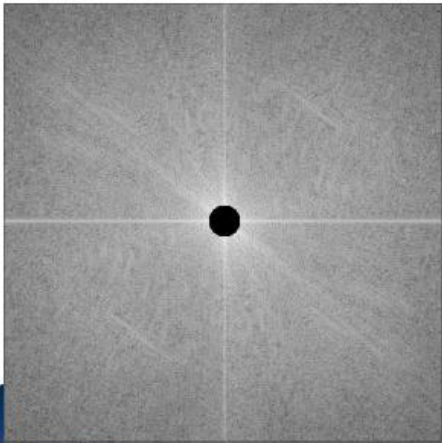


$\mathcal{F}$   
→

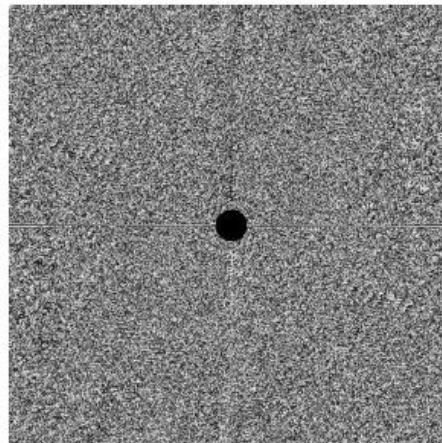
$T(l,m)$



$V(u,v)$  amplitude



$V(u,v)$  phase

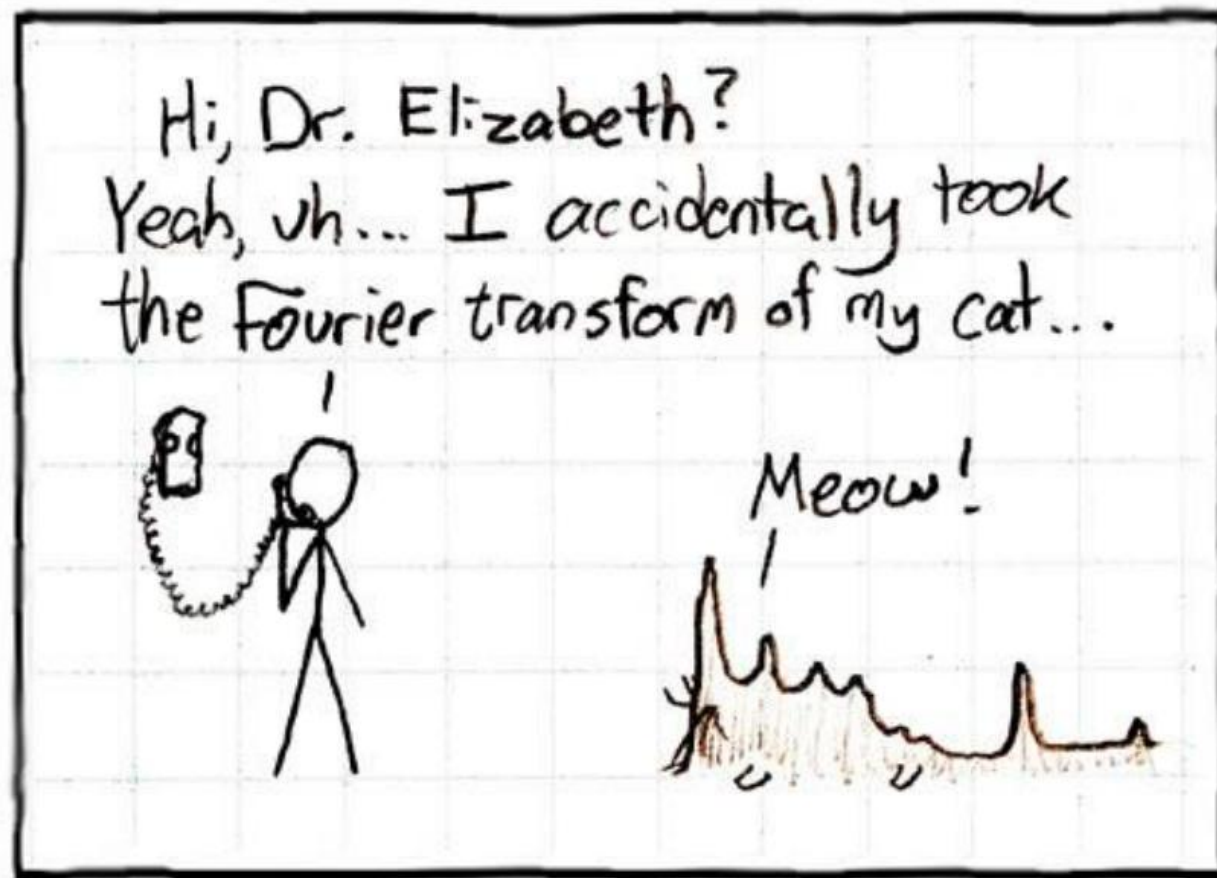


$\mathcal{F}$   
→

$T(l,m)$



[xkcd.com/26/](http://xkcd.com/26/)



# Formal Description of Imaging

$$V(u, v) \xrightarrow{\mathcal{F}} T(l, m)$$

- sample Fourier domain at discrete points  $S(u, v) = \sum_{k=1}^M \delta(u - u_k, v - v_k)$
- Fourier transform sampled visibility function  $V(u, v)S(u, v) \xrightarrow{\mathcal{F}} T^D(l, m)$
- apply the convolution theorem  $T(l, m) * s(l, m) = T^D(l, m)$

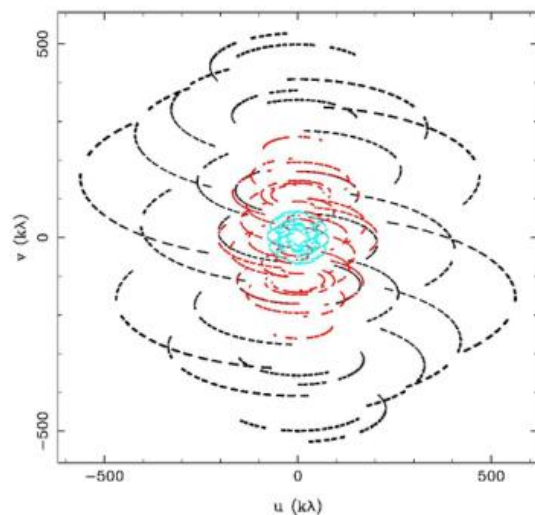
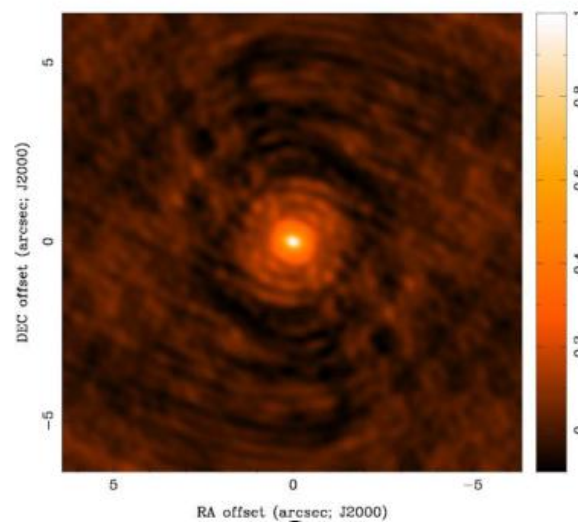
where the Fourier transform of the sampling pattern  $s(l, m) \xrightarrow{\mathcal{F}} S(u, v)$  is the “point spread function”

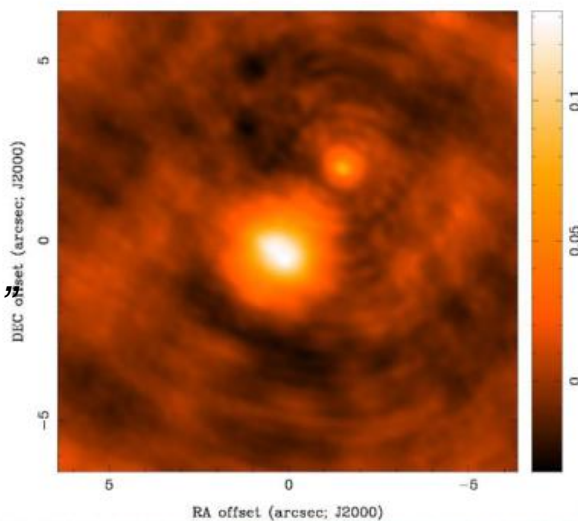
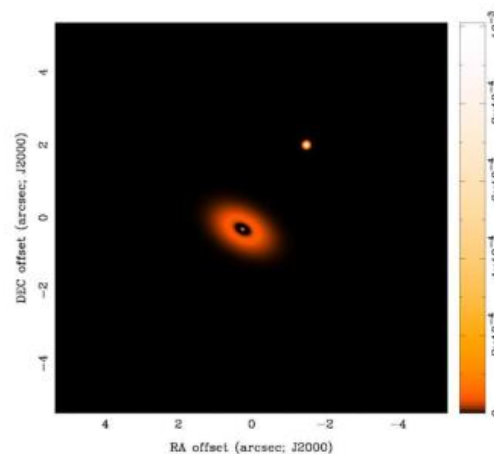
*the Fourier transform of the sampled visibilities yields the true sky brightness convolved with the point spread function*

radio jargon: the “dirty image” is the true image convolved with the “dirty beam”



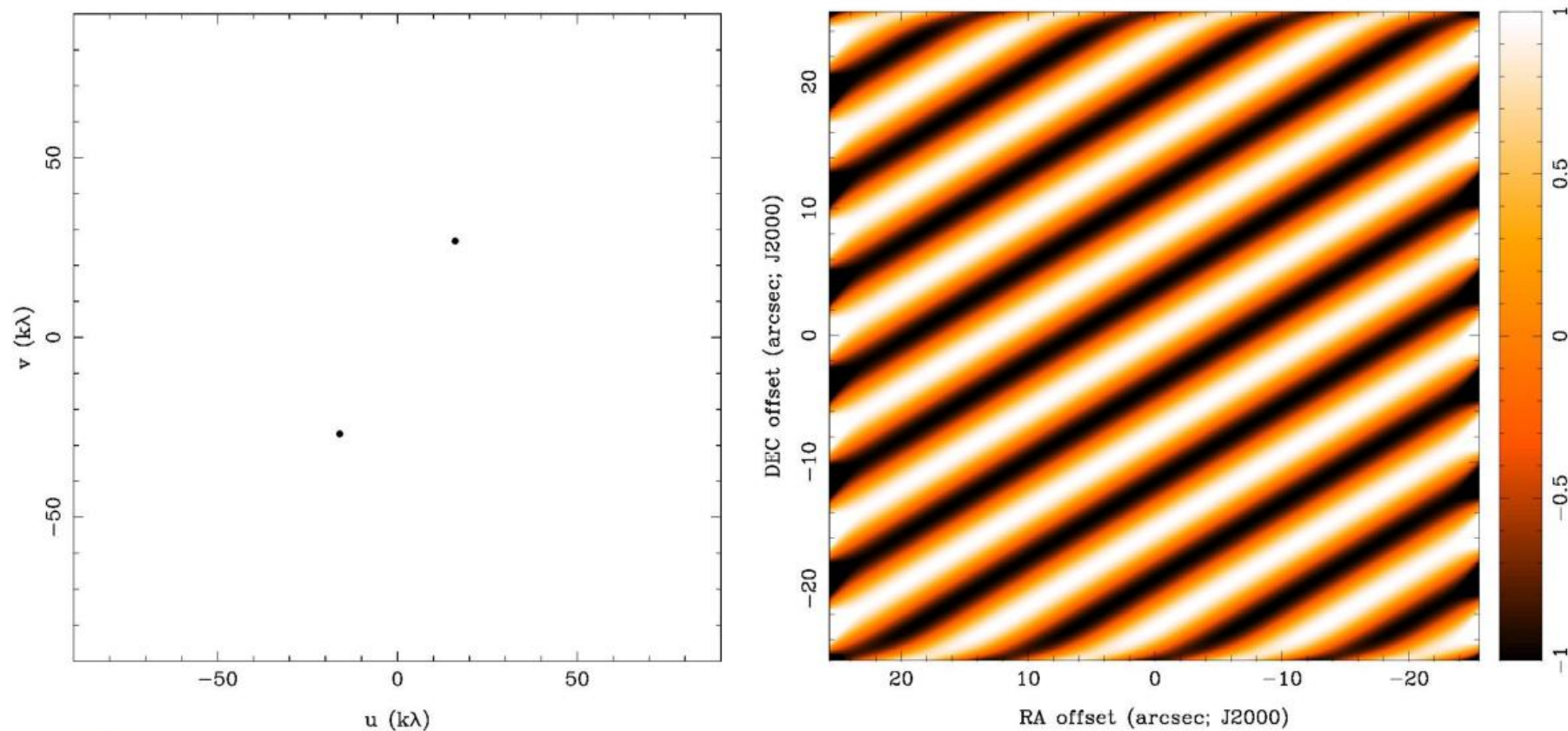
# Dirty Beam and Dirty Image

 $S(u, v)$ 

 $\mathcal{F}$ 

 $s(l, m)$   
 “dirty beam”

 $*$ 
 $T^D(l, m)$   
 “dirty image”

 $\mathcal{F}$ 

 $T(l, m)$ 

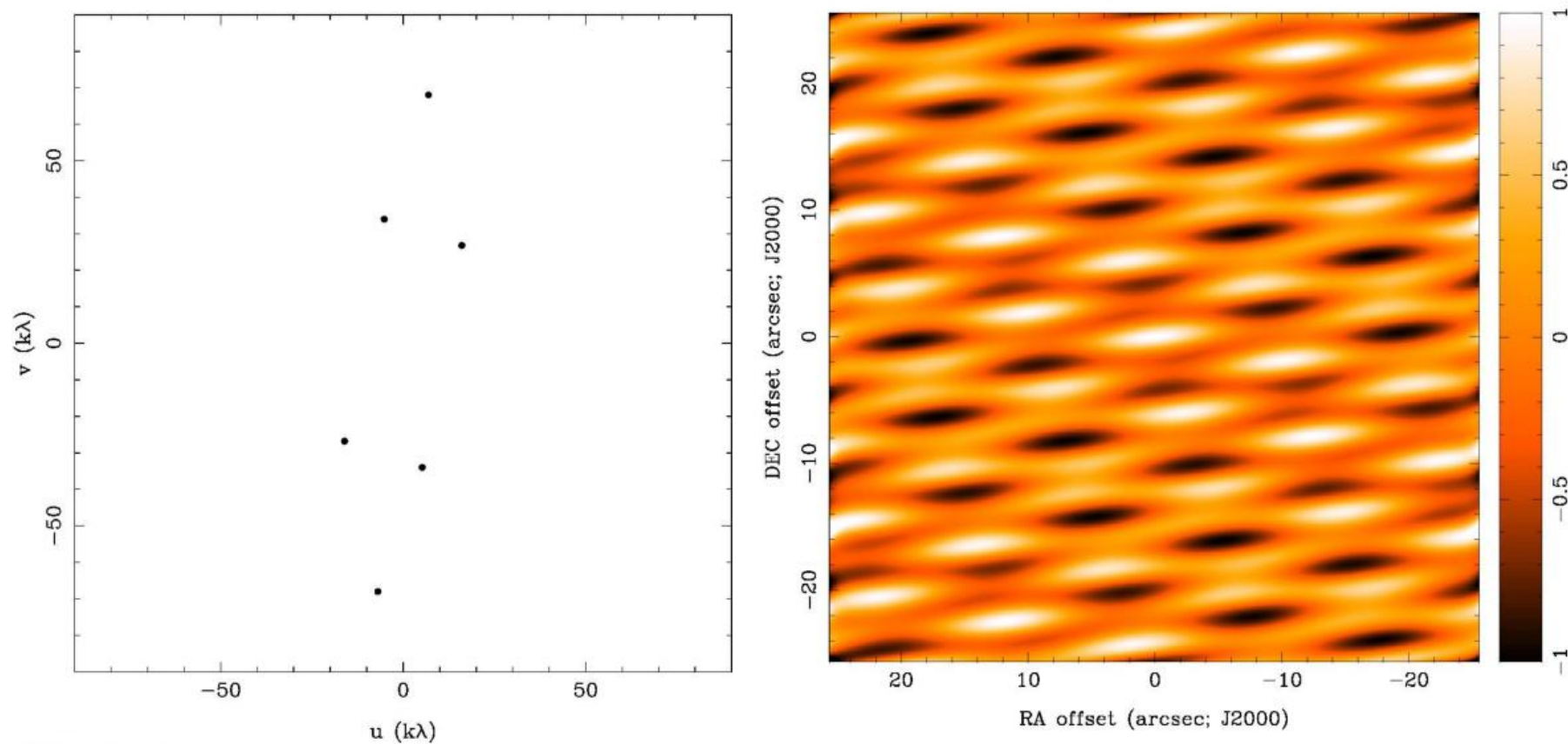

# Dirty Beam Shape and N Antennas

2 Antennas, 1 Sample



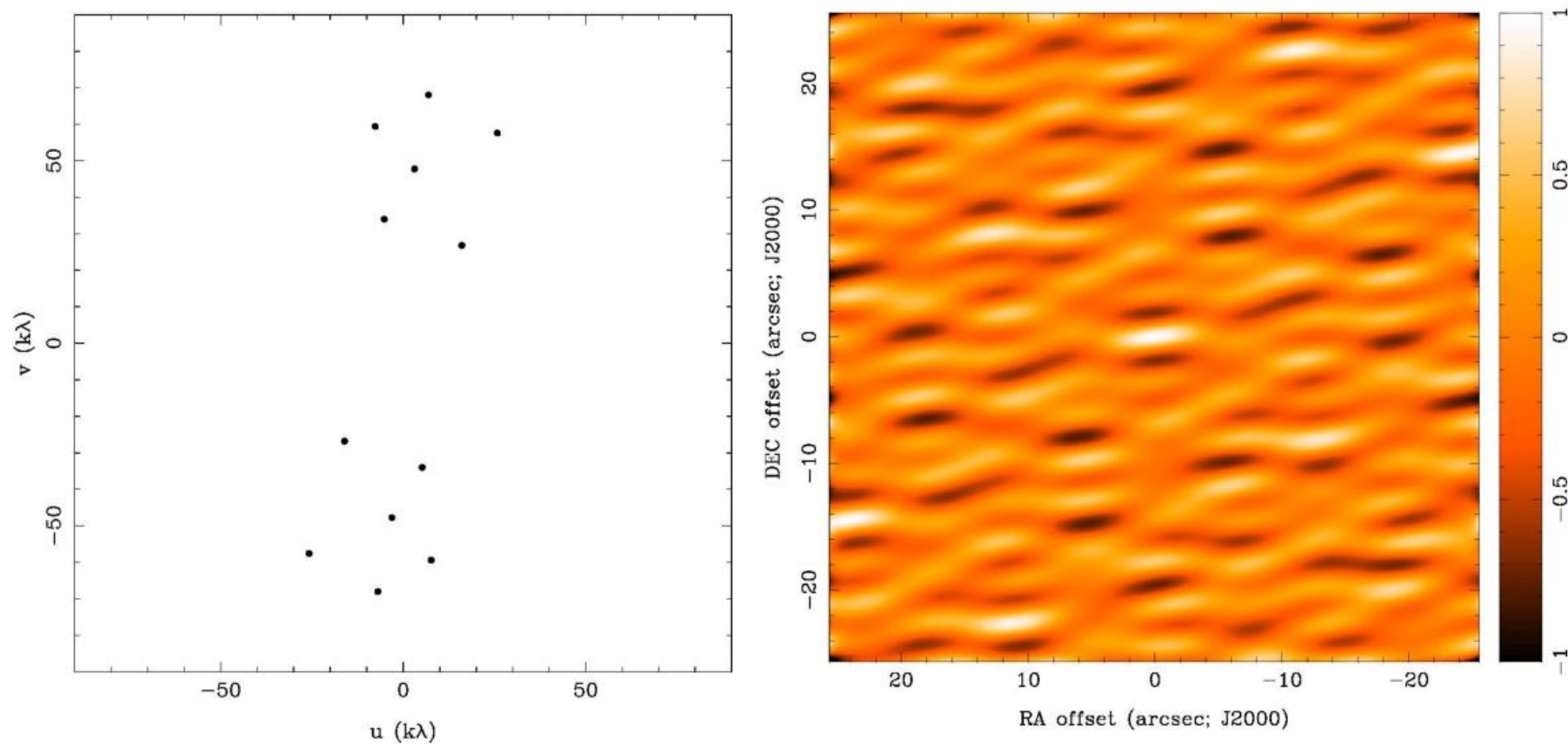
# Dirty Beam Shape and N Antennas

3 Antennas, 1 Sample



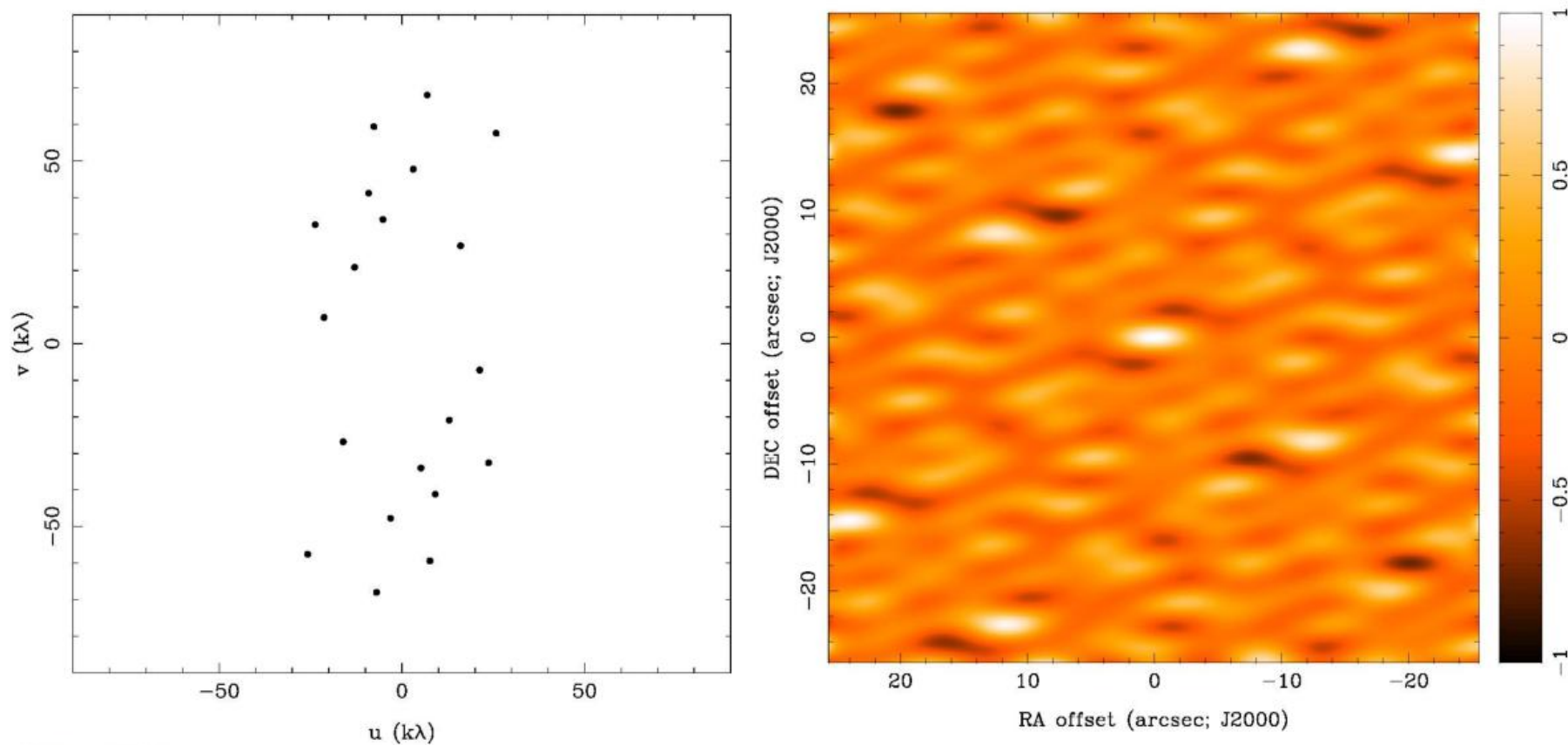
# Dirty Beam Shape and N Antennas

4 Antennas, 1 Sample



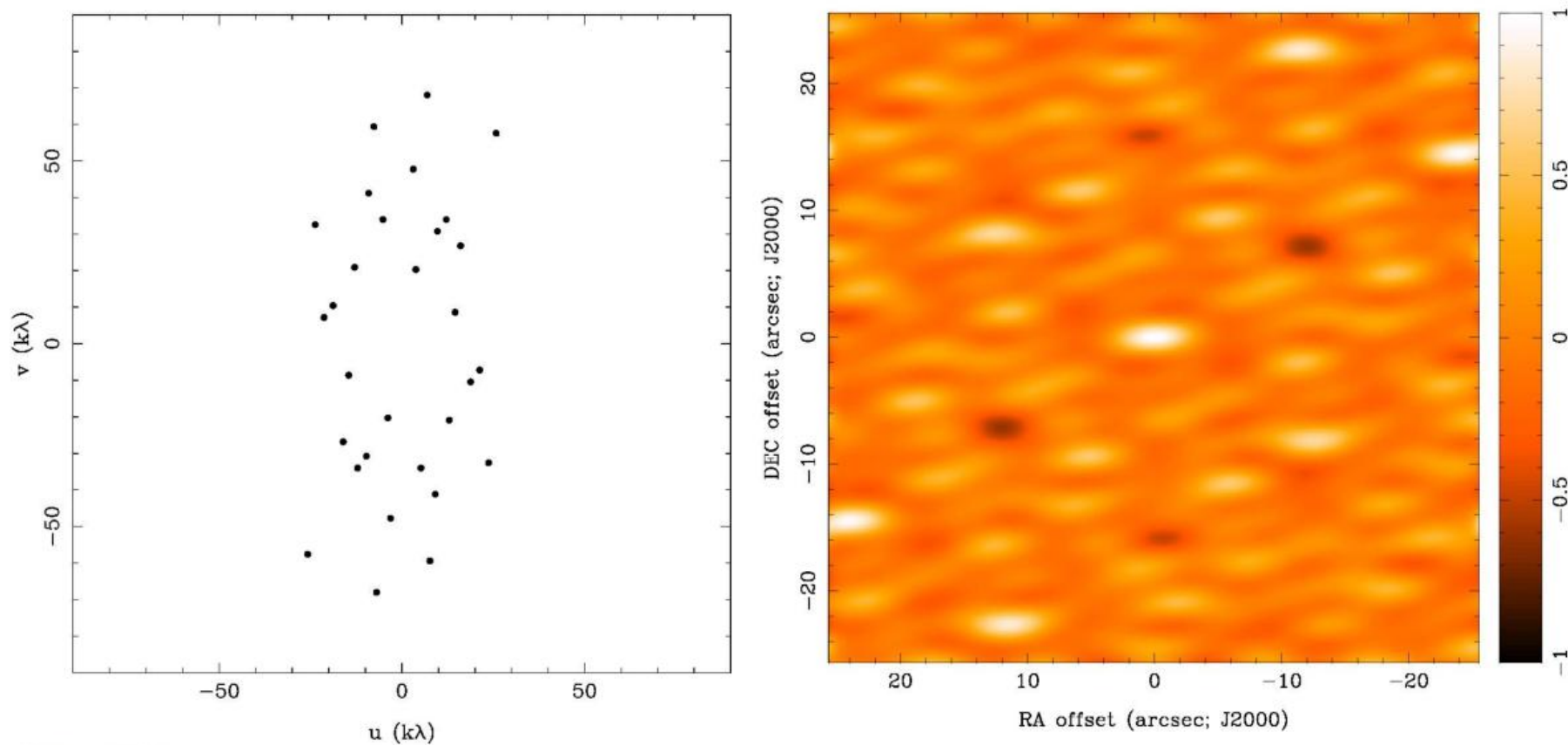
# Dirty Beam Shape and N Antennas

5 Antennas, 1 Sample



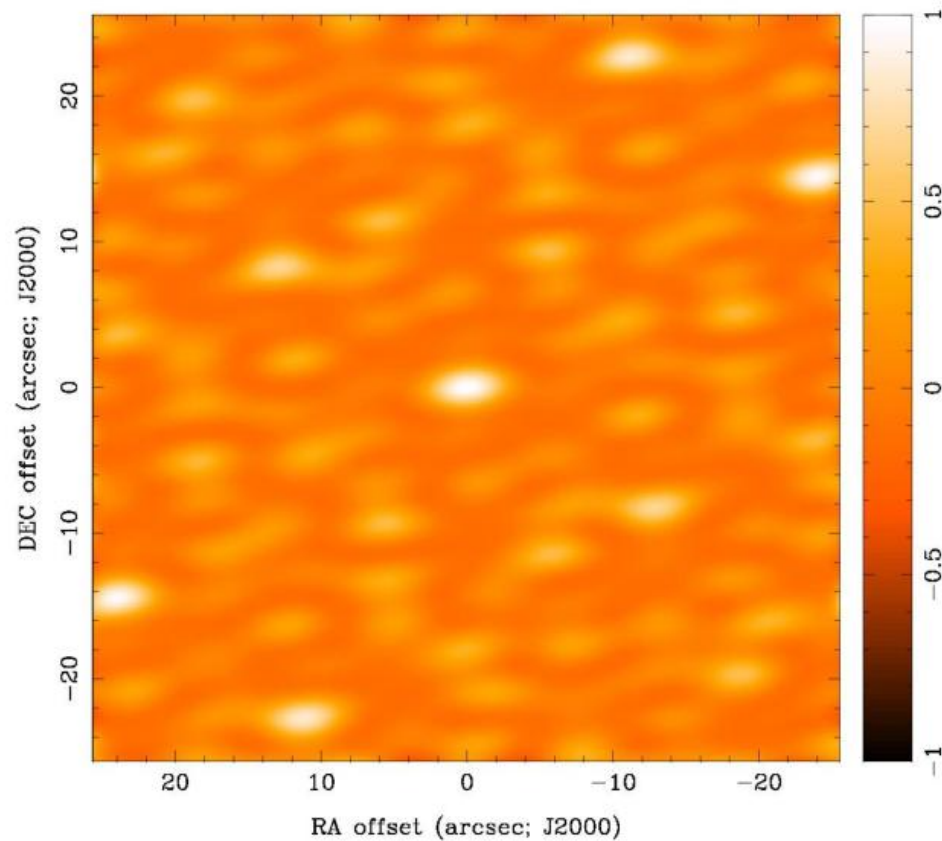
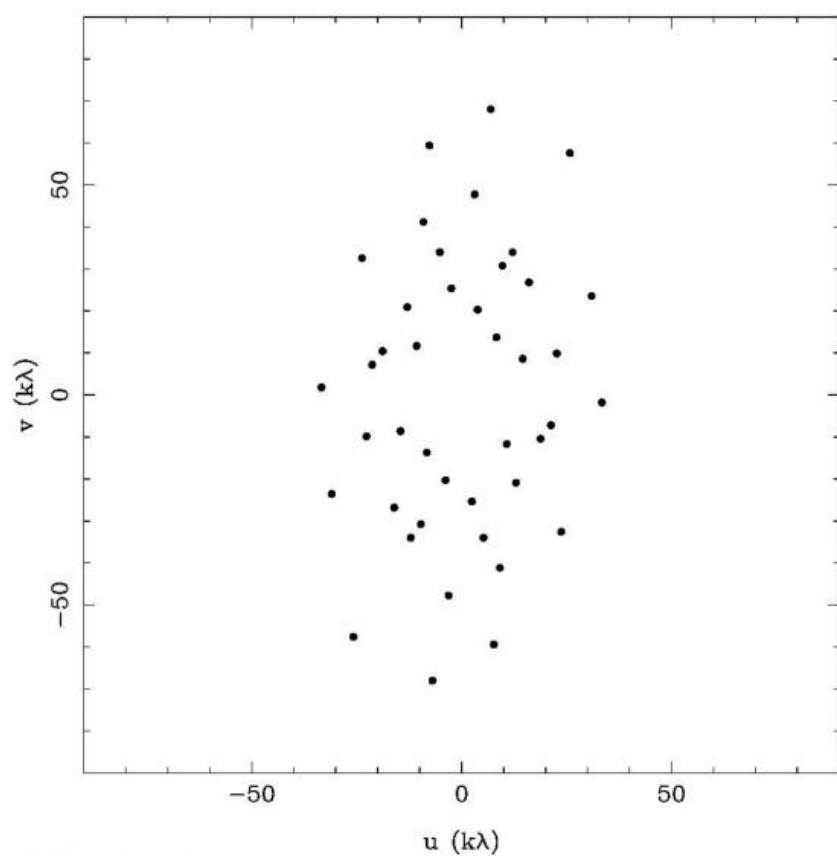
# Dirty Beam Shape and N Antennas

6 Antennas, 1 Sample



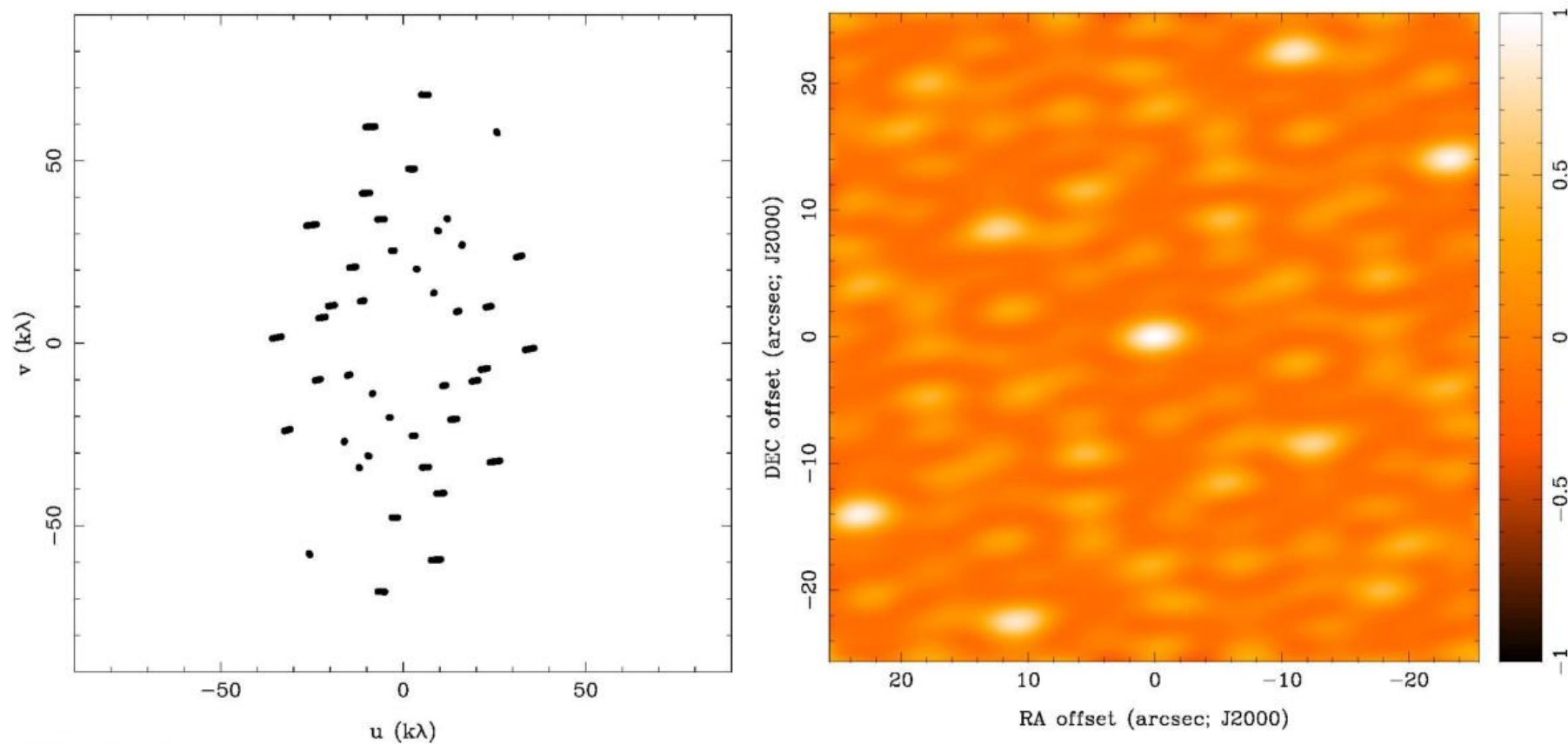
# Dirty Beam Shape and N Antennas

7 Antennas, 1 Sample



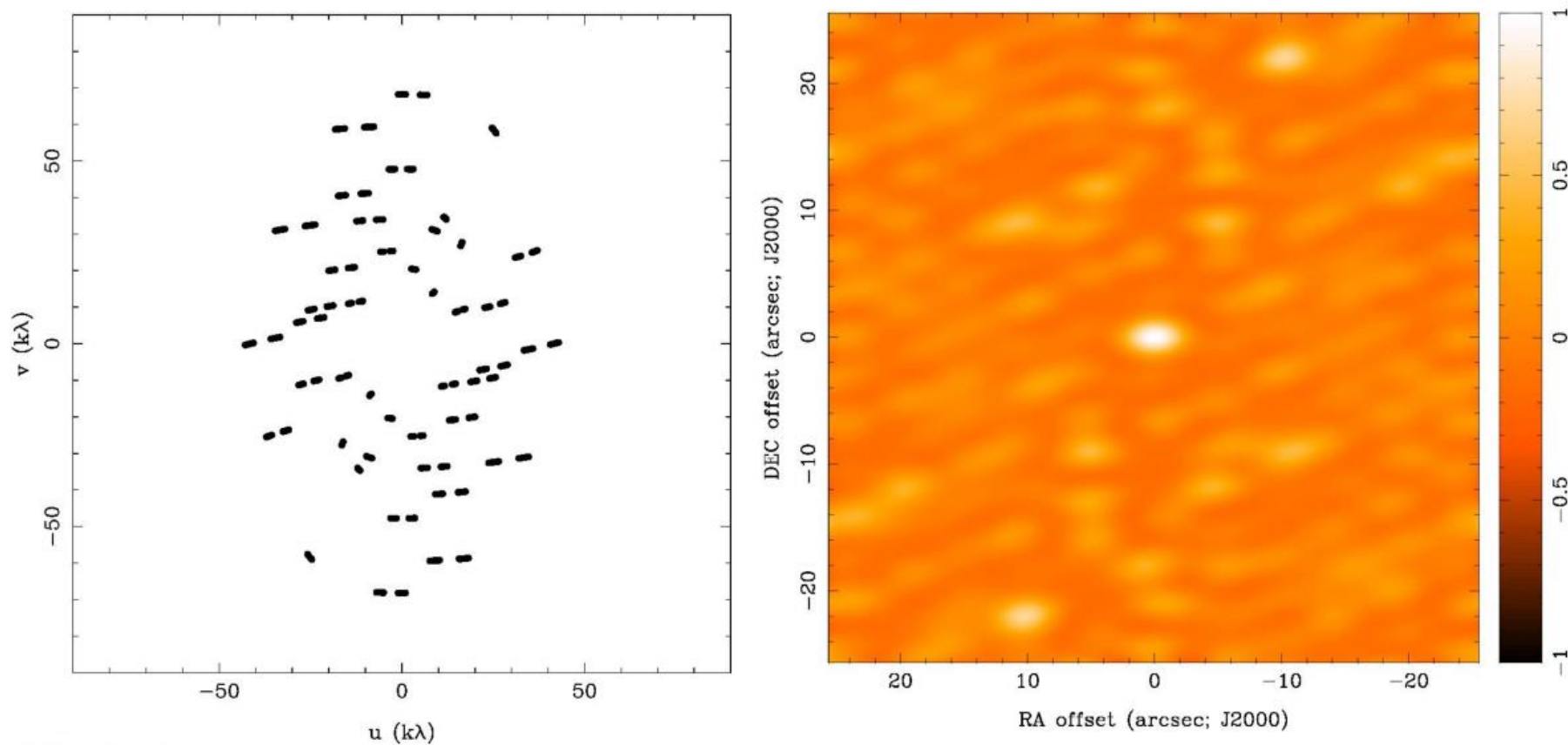
# Dirty Beam Shape and N Antennas

7 Antennas, 10 min



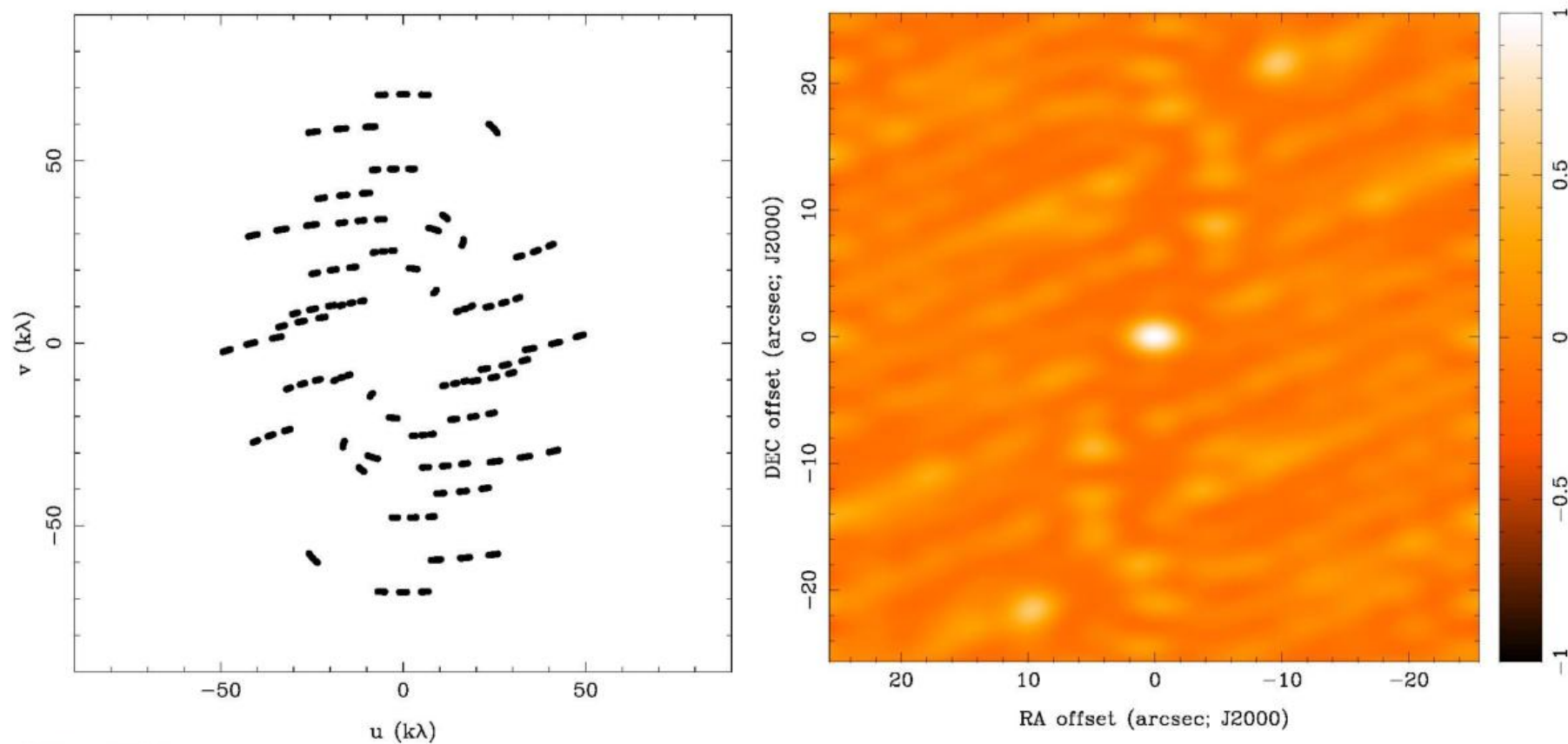
# Dirty Beam Shape and N Antennas

7 Antennas, 2 x 10 min



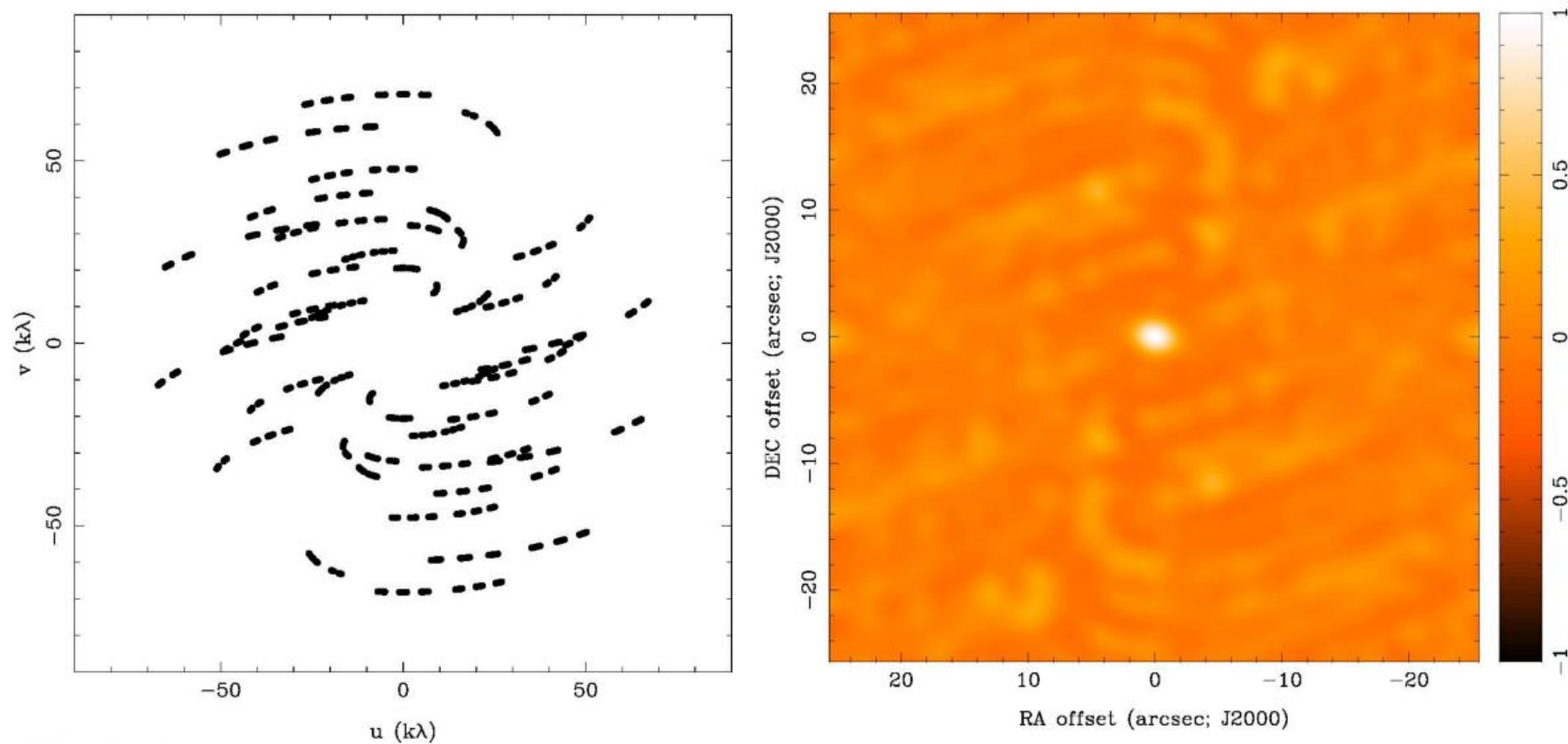
# Dirty Beam Shape and N Antennas

7 Antennas, 1 hour



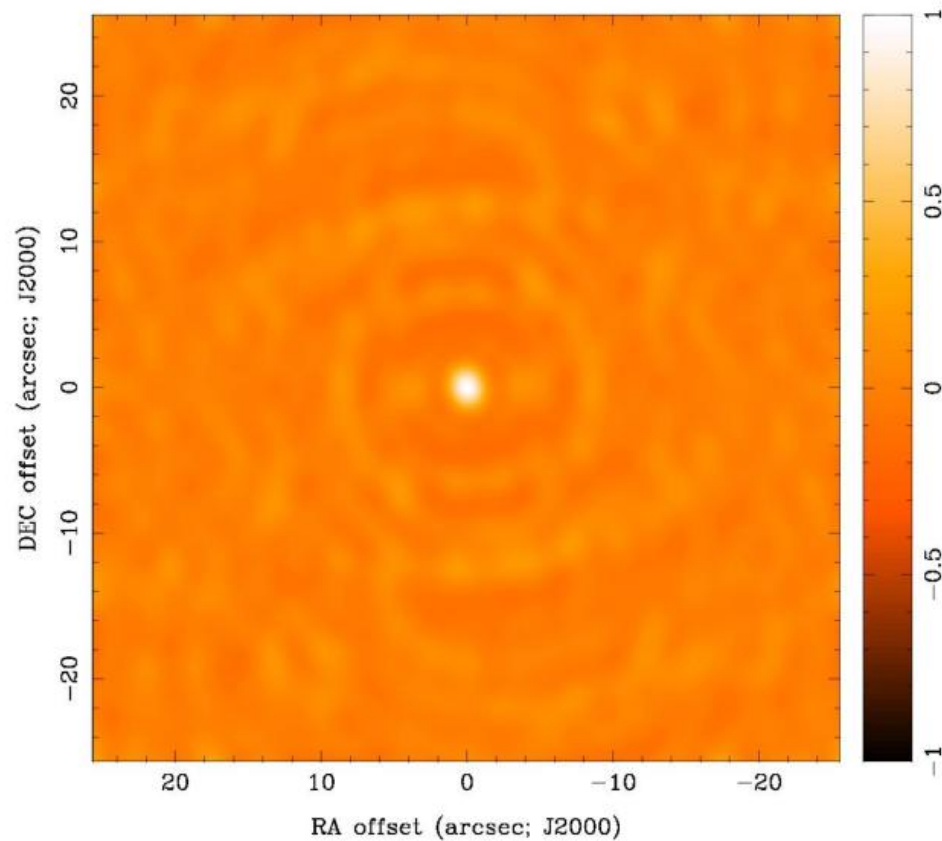
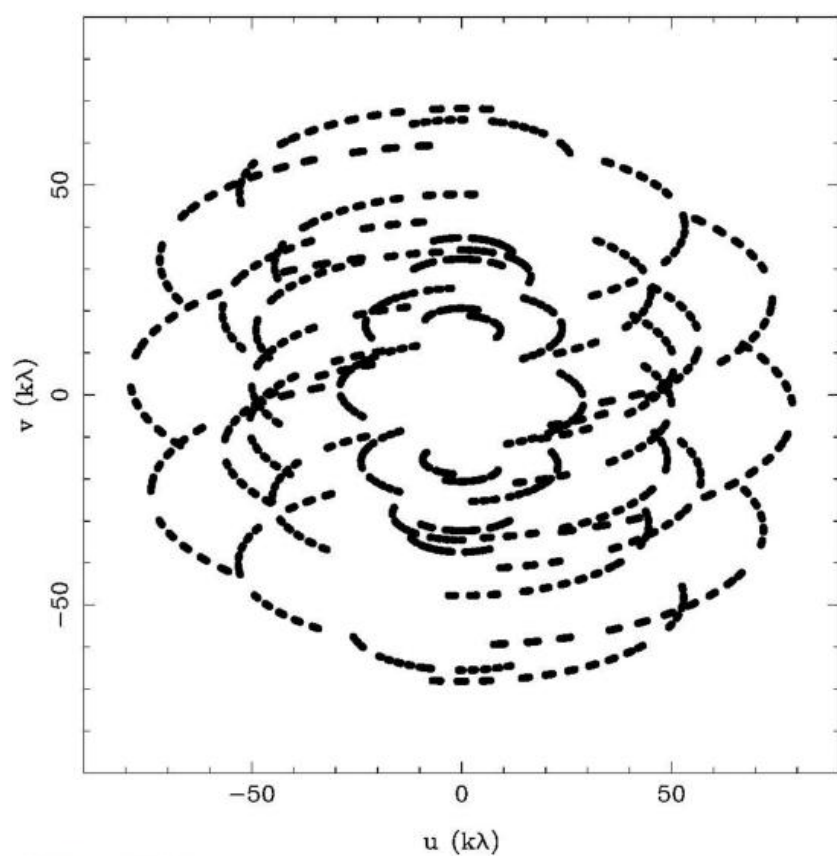
# Dirty Beam Shape and N Antennas

7 Antennas, 3 hours



# Dirty Beam Shape and N Antennas

7 Antennas, 8 hours



# Calibrated Visibilities: What's Next?

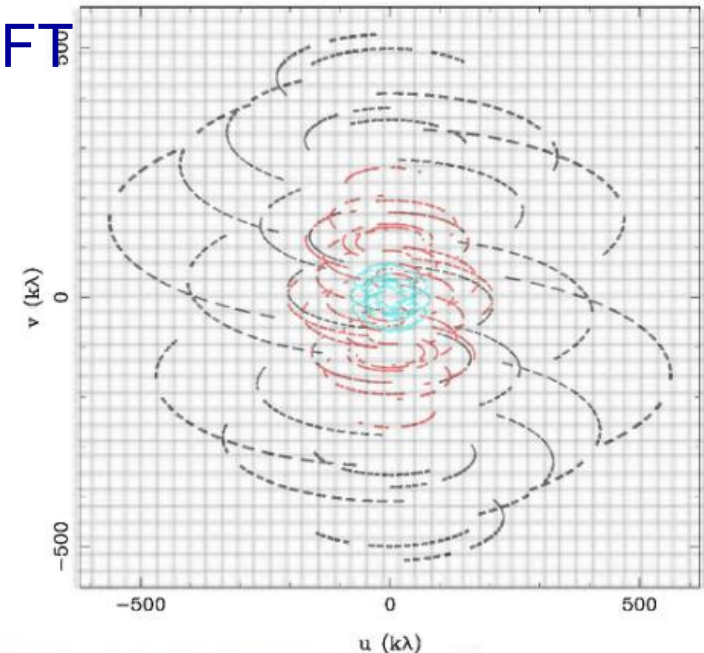
- analyze directly  $V(u,v)$  samples by model fitting
  - good for simple structures, e.g. point sources, symmetric disks
  - sometimes for statistical descriptions of sky brightness
  - visibilities have very well defined noise properties
- recover an image from the observed incomplete and noisy samples of its Fourier transform for analysis
  - Fourier transform  $V(u,v)$  to get  $T^D(l,m)$
  - difficult to do science with the dirty image  $T^D(l,m)$
  - deconvolve  $s(l,m)$  from  $T^D(l,m)$  to determine a model of  $T(l,m)$
  - work with the model of  $T(l,m)$



# Some Details of the Dirty Image

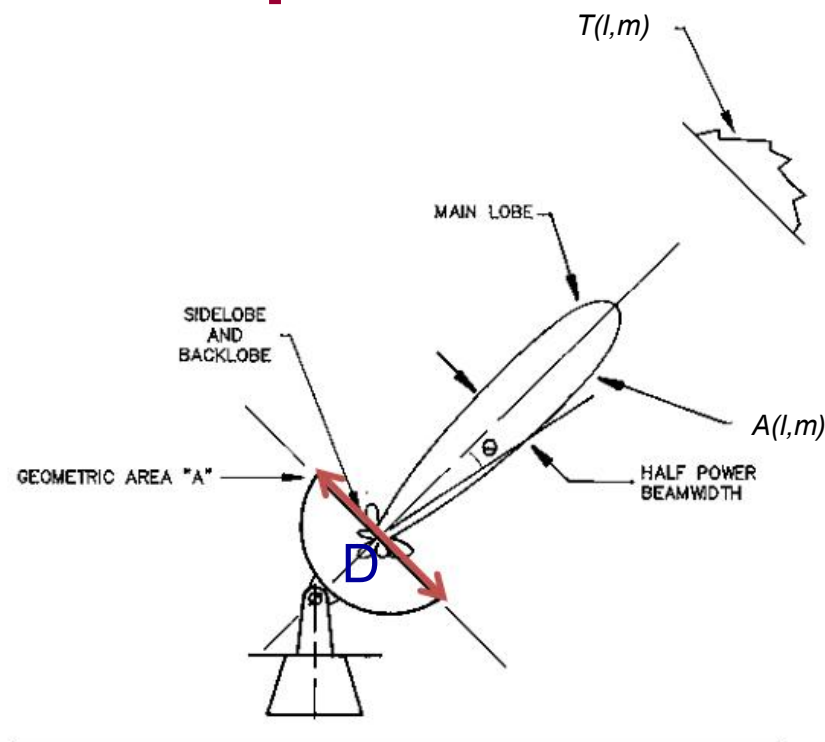
- “Fourier transform”
  - Fast Fourier Transform (FFT) algorithm is much faster than simple Fourier summation,  $O(N \log N)$  for  $2^N \times 2^N$  image
  - FFT requires data on a regularly spaced grid
  - aperture synthesis does not provide  $V(u, v)$  on a regularly spaced grid, so...
- “gridding” used to resample  $V(u, v)$  for FFT
  - customary to use a convolution method
  - special (“spheroidal”) functions that minimize smoothing and aliasing

$$V^G(u, v) = V(u, v)S(u, v) * G(u, v) \\ \xrightarrow{F} T^D(l, m)g(l, m)$$



# Antenna Primary Beam Response

- antenna response  $A(l,m)$  is not uniform across the entire sky
  - main lobe = “primary beam”  
fwhm  $\sim \lambda/D$
  - response beyond primary beam can be important (“sidelobes”)
- antenna beam modifies the sky brightness distribution
  - $T(l,m) \rightarrow T(l,m)A(l,m)$
  - can correct with division by  $A(l,m)$  in the image plane
  - large source extents require multiple pointings of antennas = mosaicking



SMA 6 m  
345 GHz

ALMA 12 m  
690 GHz



# Imaging Decisions: Pixel Size, Image Size

- pixel size
  - satisfy sampling theorem for longest baselines

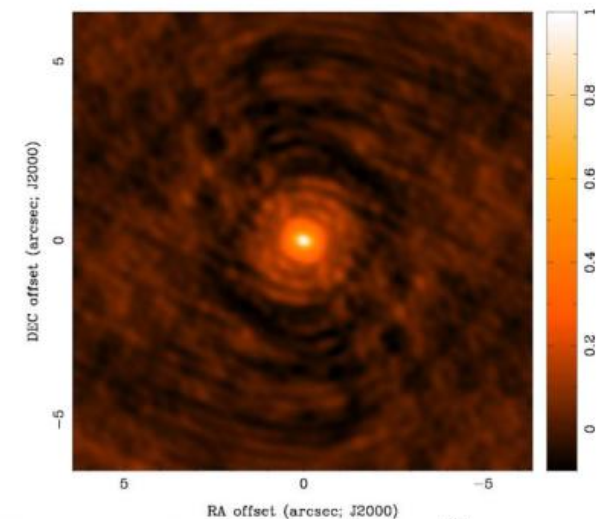
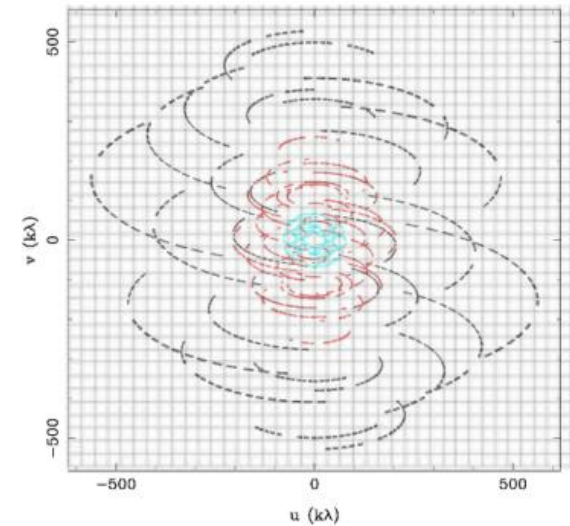
$$\Delta l < \frac{1}{2u_{max}} \quad \Delta m < \frac{1}{2v_{max}}$$

- in practice, 3 to 5 pixels across main lobe of dirty beam to aid deconvolution
  - e.g. at 870  $\mu\text{m}$  with baselines to 500 meters  $\rightarrow$  pixel size  $< 0.1$  arcsec
  - CASA “cell” size
- image size
  - natural choice is often the full extent of the primary beam  $A(l,m)$
  - e.g. SMA at 870  $\mu\text{m}$ , 6 meter antennas  $\rightarrow$  image size  $2 \times 35$  arcsec
  - if there are bright sources in the sidelobes of  $A(l,m)$ , then the FFT will alias them into the image  $\rightarrow$  make a larger image (or equivalent)
  - CASA “imsize”



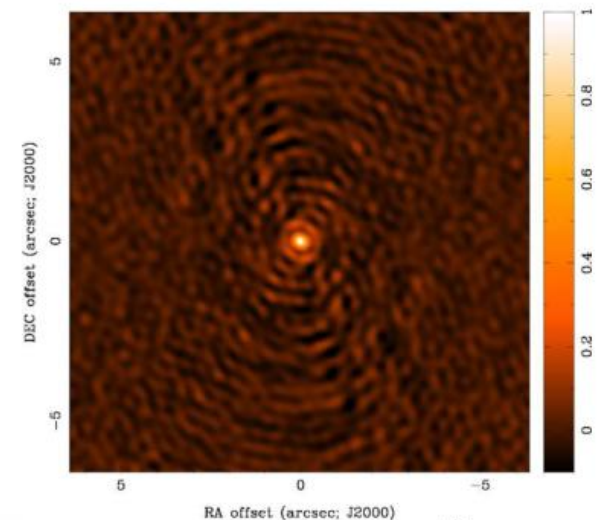
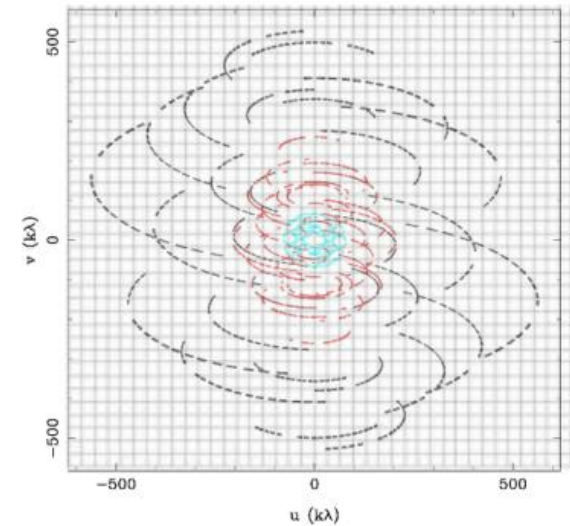
# Imaging Decisions: Visibility Weighting

- introduce weighting function  $W(u,v)$ 
  - modifies sampling function
  - $S(u,v) \rightarrow S(u,v)W(u,v)$
  - changes  $s(l,m)$ , the dirty beam shape
- natural weight
  - $W(u,v) = 1/\sigma^2$  in occupied  $(u,v)$  cells, where  $\sigma^2$  is the noise variance, and  $W(u,v) = 0$  everywhere else
  - maximizes point source sensitivity
  - lowest rms in image
  - generally gives more weight to short baselines (low spatial frequencies), so angular resolution is degraded



# Dirty Beam Shape and Weighting

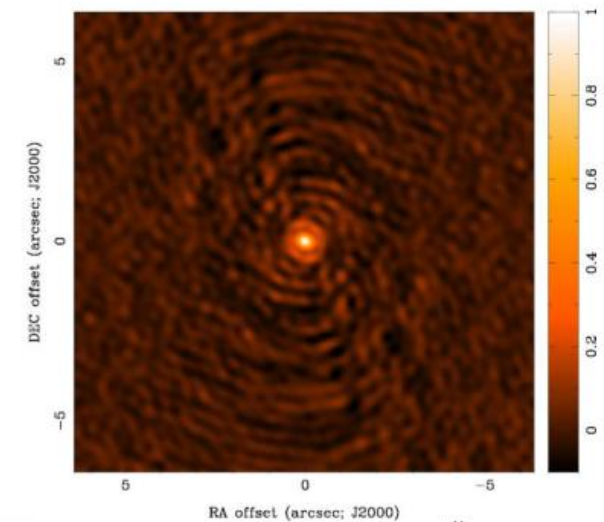
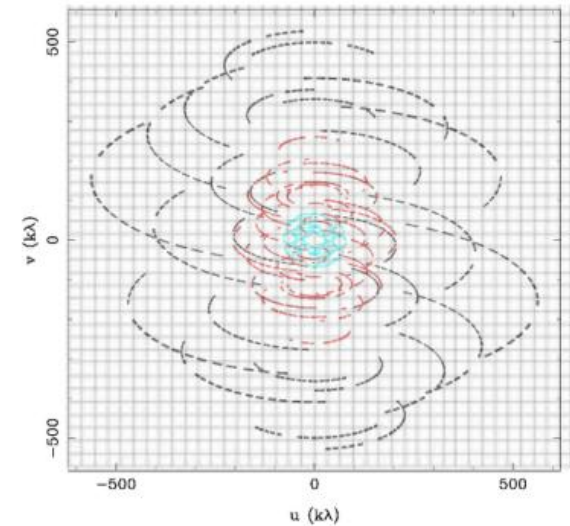
- uniform weight
  - $W(u,v)$  is inversely proportional to local density of  $(u,v)$  points
  - sum of weights in a  $(u,v)$  cell = const (and 0 for empty cells)
  - fills  $(u,v)$  plane more uniformly and dirty beam sidelobes are lower
  - gives more weight to long baselines (high spatial frequencies), so angular resolution is enhanced
  - downweights some data, so point source sensitivity is degraded
  - can be trouble with sparse sampling: cells with few data points have same weight as cells with many data points



# Dirty Beam Shape and Weighting

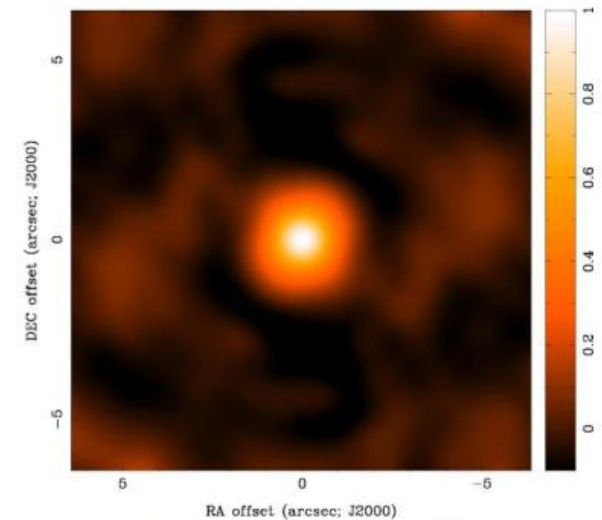
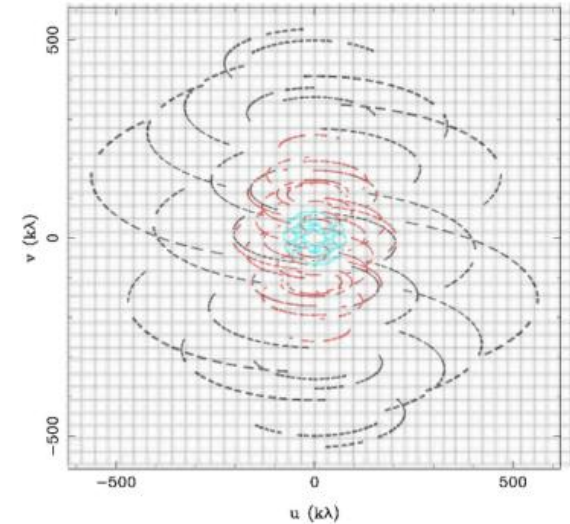
- robust (Briggs) weight
  - variant of uniform that avoids giving too much weight to  $(u,v)$  cells with low natural weight
  - software implementations differ
  - e.g. 
$$W(u, v) = \frac{1}{\sqrt{1 + S_N^2 / S_{thresh}^2}}$$

$S_N$  is natural weight of cell  
 $S_{thresh}$  is a threshold  
 high threshold  $\rightarrow$  natural weight  
 low threshold  $\rightarrow$  uniform weight
- *an adjustable parameter allows for continuous variation between maximum point source sensitivity and resolution*



# Dirty Beam Shape and Weighting

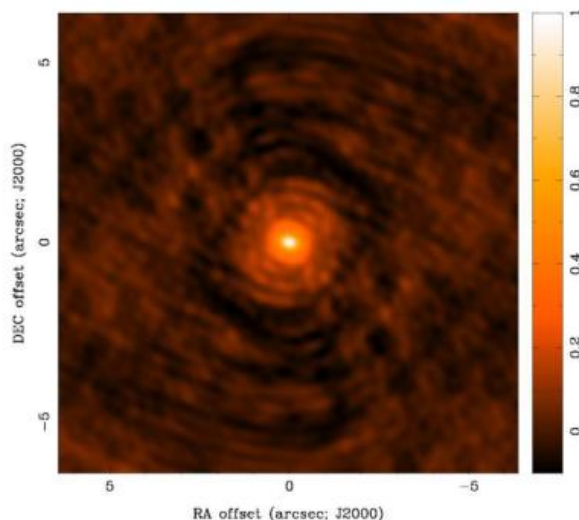
- tapering
    - apodize  $(u, v)$  sampling by a Gaussian
- $$W(u, v) = \exp\left(-\frac{(u^2 + v^2)}{t^2}\right)$$
- $t$  = adjustable tapering parameter
- like smoothing in the image plane (convolution by a Gaussian)
  - gives more weight to short baselines, degrades angular resolution
  - downweights some data, so point source sensitivity degraded
  - may improve sensitivity to extended structure sampled by short baselines
  - limits to usefulness



# Weighting and Tapering: Image Noise

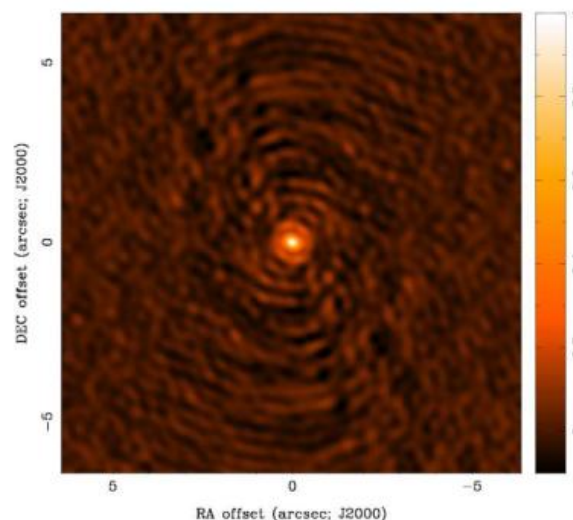
natural  
0.59x0.50

rms=1.0



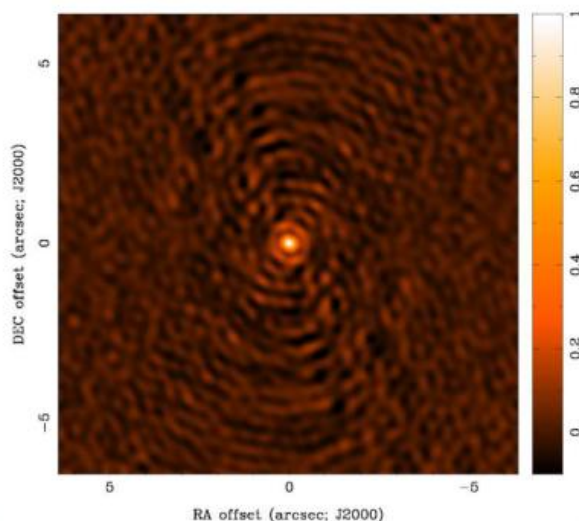
robust=0  
0.40x0.34

rms=1.3



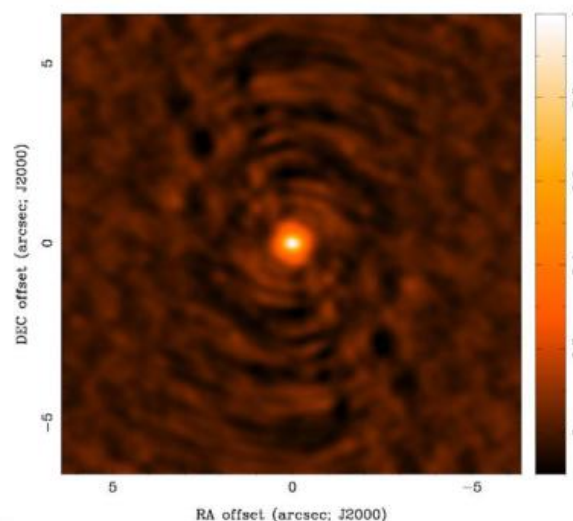
uniform  
0.35x0.30

rms=2.1



robust=0  
+ taper to  
0.59x0.50

rms=1.2



# Weighting and Tapering: Summary

- imaging parameters provide a lot of freedom
- appropriate choices depend on science goals

|                             | Robust/Uniform | Natural | Taper   |
|-----------------------------|----------------|---------|---------|
| resolution                  | higher         | medium  | lower   |
| sidelobes                   | lower          | higher  | depends |
| point source sensitivity    | lower          | maximum | lower   |
| extended source sensitivity | lower          | medium  | higher  |



# Beyond the Dirty Image: Deconvolution

- to keep you awake at night
  - $\exists$  an infinite number of  $T(l,m)$  compatible with sampled  $V(u,v)$ , with “invisible” distributions  $R(l,m)$  where  $s(l,m) * R(l,m) = 0$ 
    - no data beyond  $u_{\max}, v_{\max} \rightarrow$  unresolved structure
    - no data within  $u_{\min}, v_{\min} \rightarrow$  limit on largest size scale
    - holes in between  $\rightarrow$  synthesized beam sidelobes
  - noise  $\rightarrow$  undetected/corrupted structure in  $T(l,m)$
  - no unique prescription for extracting optimum estimate of  $T(l,m)$
- deconvolution
  - uses non-linear techniques to interpolate/extrapolate samples of  $V(u,v)$  into unsampled regions of the  $(u,v)$  plane
  - aims to find a sensible model of  $T(l,m)$  compatible with data
  - requires *a priori* assumptions about  $T(l,m)$  to pick plausible “invisible” distributions to fill unmeasured parts of the Fourier plane



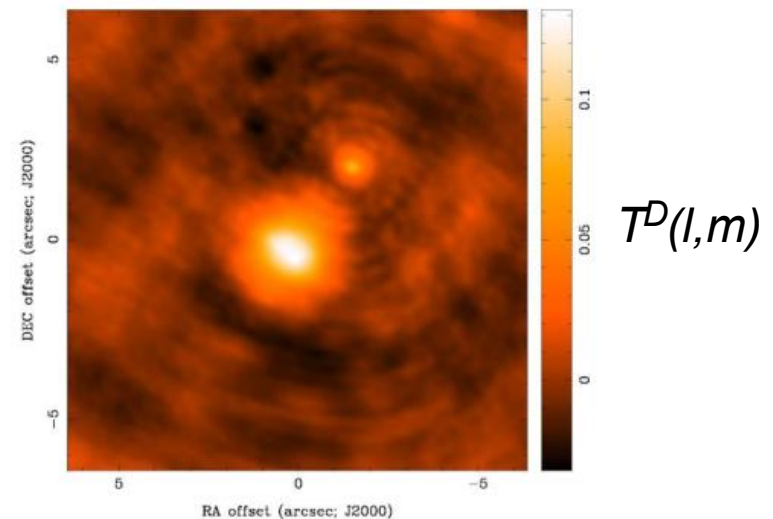
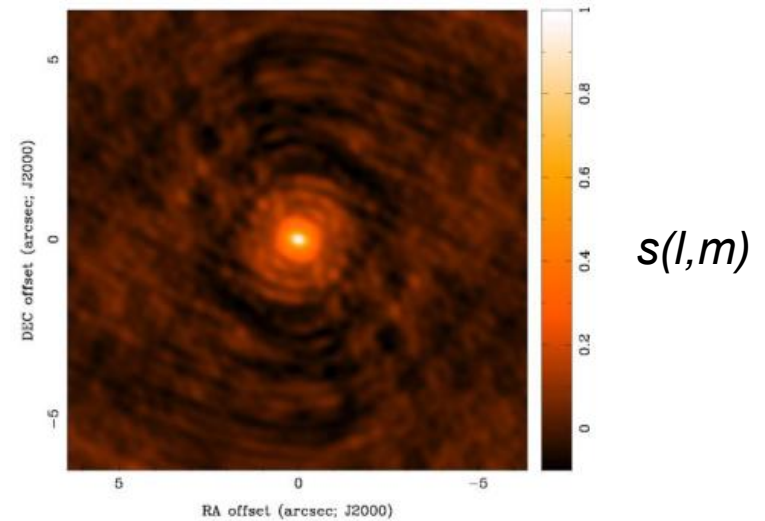
# Deconvolution Algorithms

- an active research area, e.g. compressive sensing methods
- **clean**: dominant deconvolution algorithm in radio astronomy
  - *a priori* assumption:  $T(l,m)$  is a collection of point sources
  - fit and subtract the synthesized beam iteratively
  - original version by Högbom (1974) purely image based
  - variants developed for higher computational efficiency, model visibility subtraction, to deal better with extended emission structure, etc.
- **maximum entropy**: a rarely used alternative
  - *a priori* assumption:  $T(l,m)$  is smooth and positive
  - define “smoothness” via a mathematical expression for entropy, e.g. Gull and Skilling (1983), find smoothest image consistent with data
  - vast literature about the deep meaning of entropy as information content



# Basic clean Algorithm

- initialize
  - a *residual map* to the dirty map
  - a *Clean Component* list
- 1. identify the highest peak in the residual map as a point source
- 2. subtract a fraction of this peak from the *residual map* using a scaled dirty beam,  $s(l,m)$  x gain
- 3. add this point source location and amplitude to the *Clean Component* list
- 4. goto step 1 (an iteration) unless stopping criterion reached



# Basic clean Algorithm (continued)

- stopping criteria?
  - *residual map* maximum < threshold = multiple of rms (if noise limited)
  - *residual map* maximum < threshold = fraction of dirty map maximum (if dynamic range limited)
  - maximum number of *Clean Components* reached (no justification)
- loop gain?
  - good results for  $g=0.1$  to  $0.3$
  - lower values can work better for smoother emission,  $g=0.05$
- easy to include *a priori* information about where in dirty map to search for *Clean Components* (using “boxes” or “masks”)
  - very useful but potentially dangerous
- Schwarz (1978) showed that the clean algorithm is equivalent to a least squares fit of sinusoids to visibilities in the case of no noise

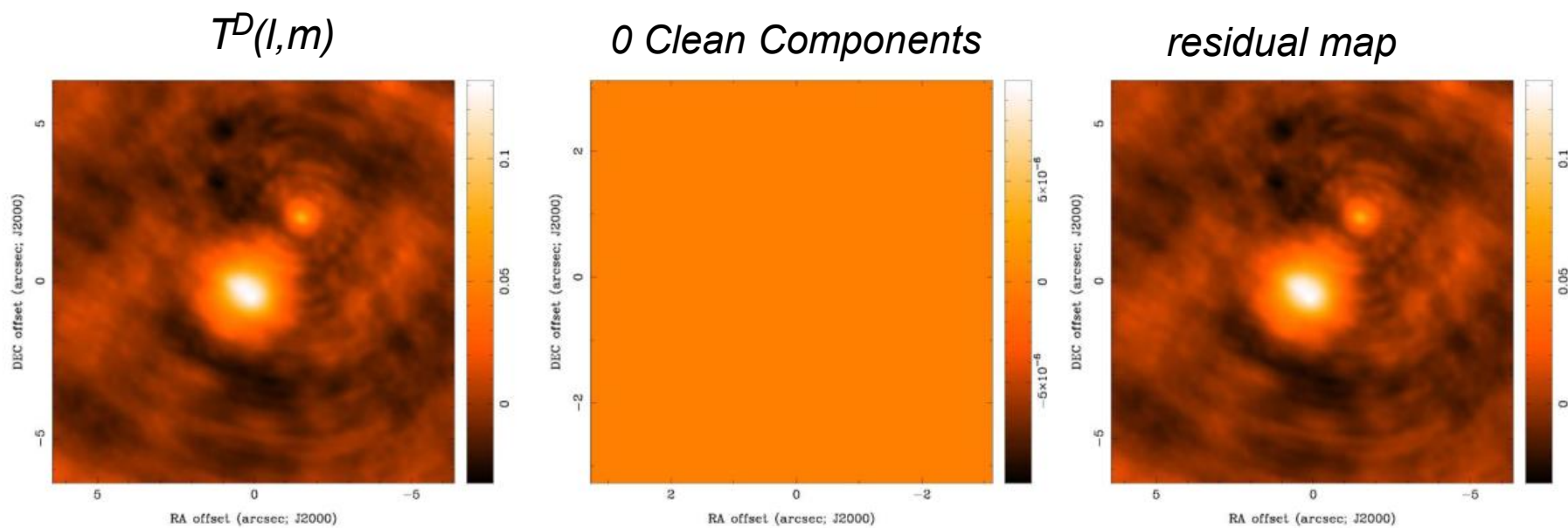


# Basic clean Algorithm (continued)

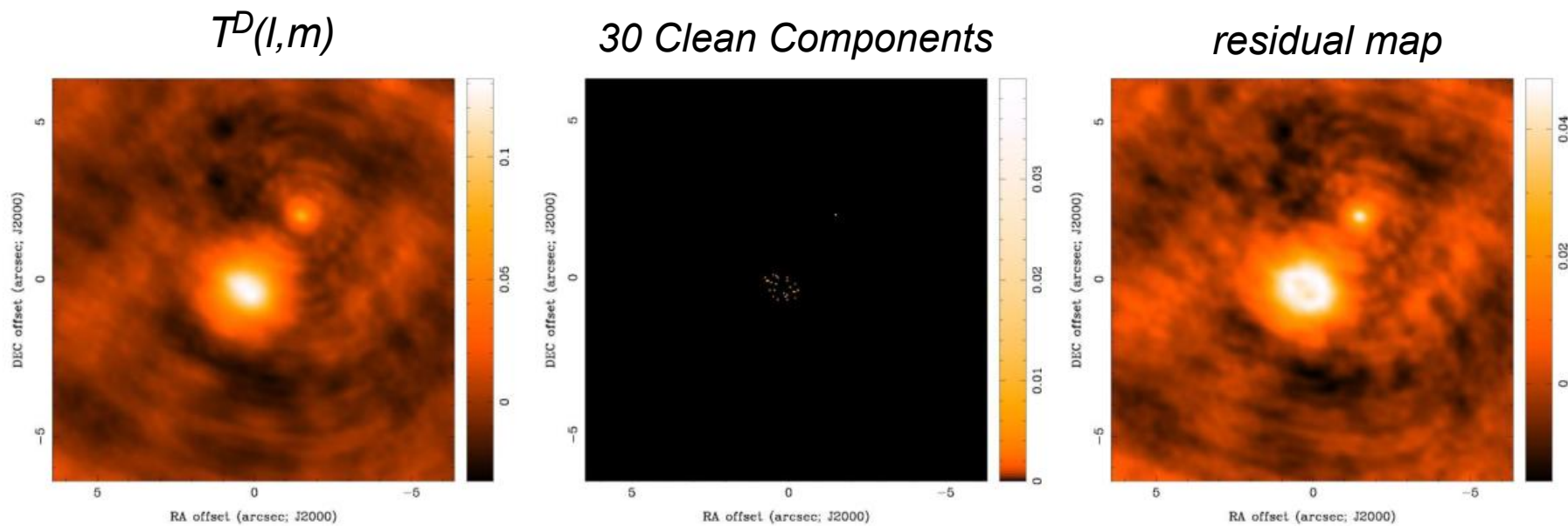
- last step: make “restored” image
  - make a model image with all point source *Clean Components*
  - convolve point sources with an elliptical Gaussian, fit to the main lobe of the dirty beam (“clean beam”); avoids super-resolution of model
  - add *residual map* of noise and source structure below the threshold
- resulting restored image is an estimate of the true sky brightness  $T(l,m)$
- units of the restored image are (mostly) Jy per clean beam area  
= intensity (or brightness temperature)
- for most weighting schemes, there is information in the image from baselines that sample high spatial frequencies within the clean beam fwhm, so modest super-resolution may be OK
- the restored image does not actually fit the observed visibilities



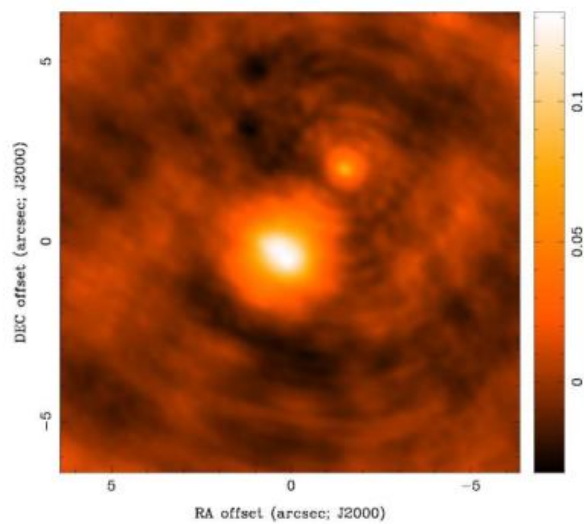
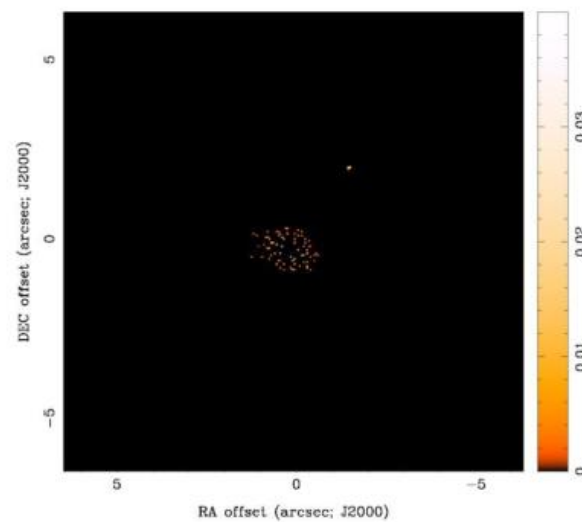
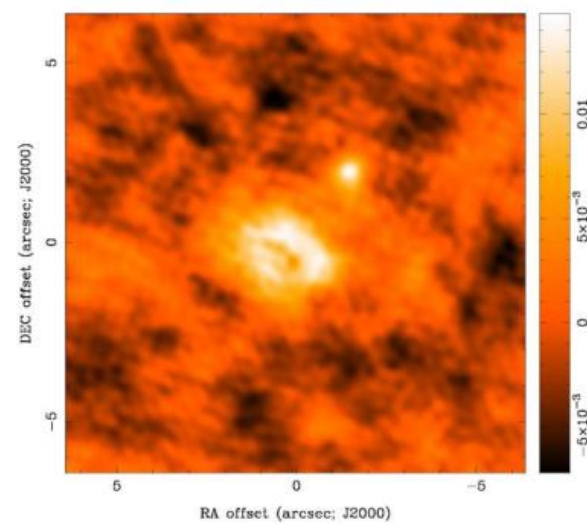
# clean Example



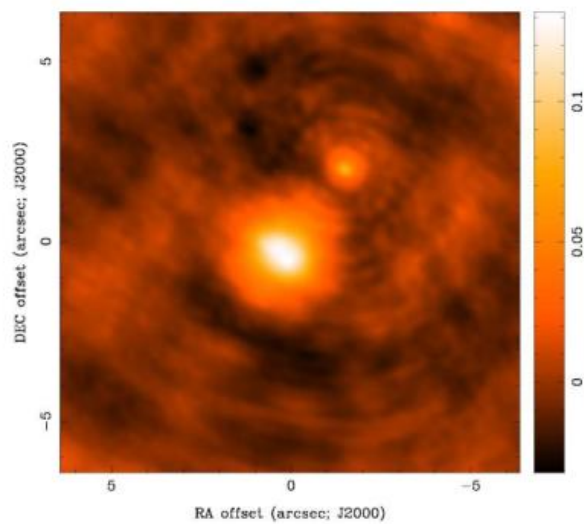
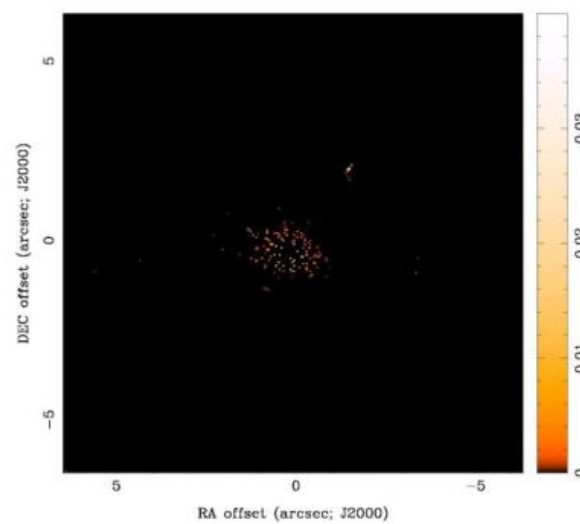
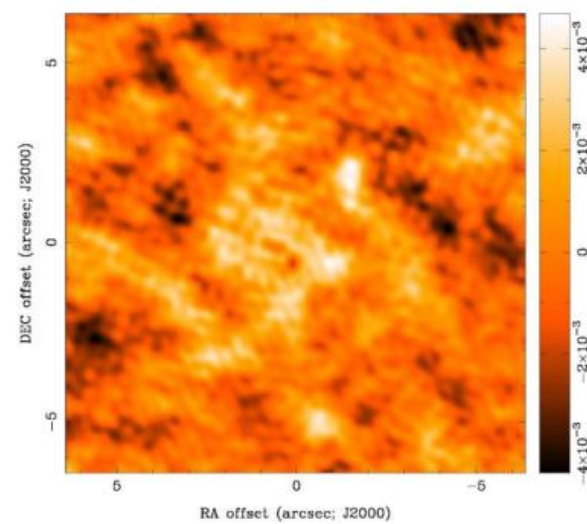
# clean Example



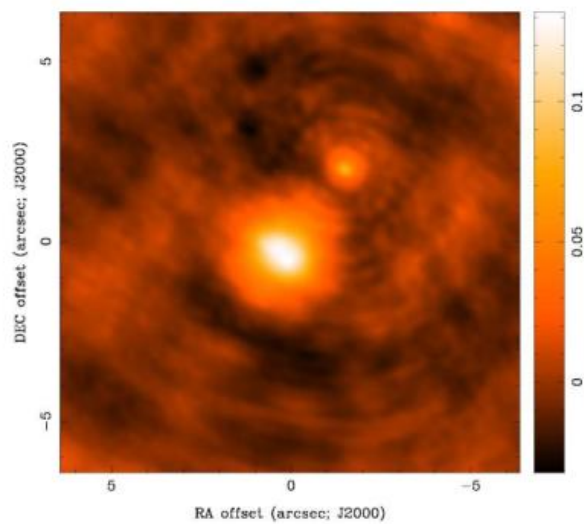
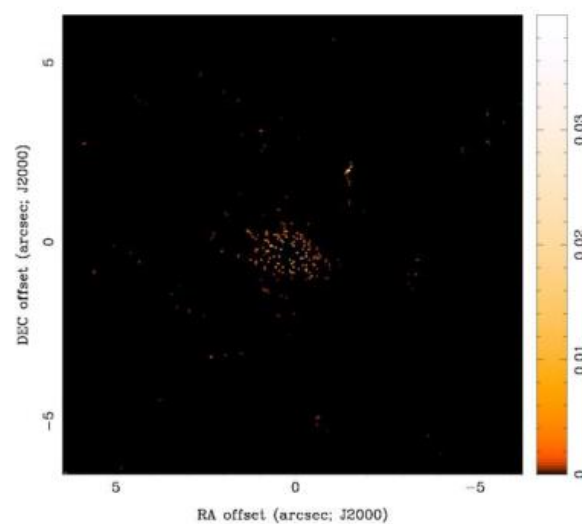
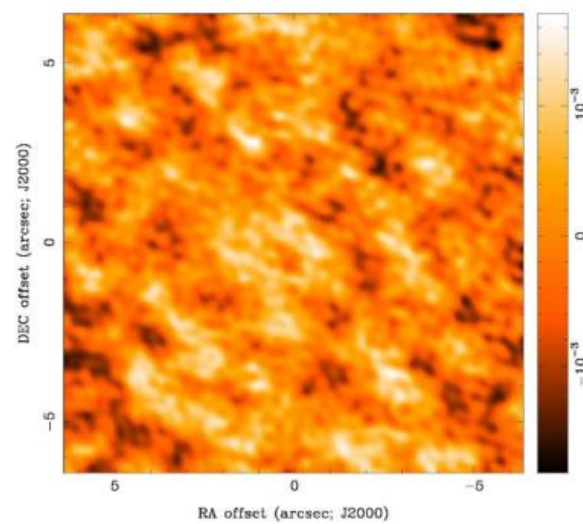
# clean Example

 $T^D(l, m)$ *100 Clean Components**residual map*

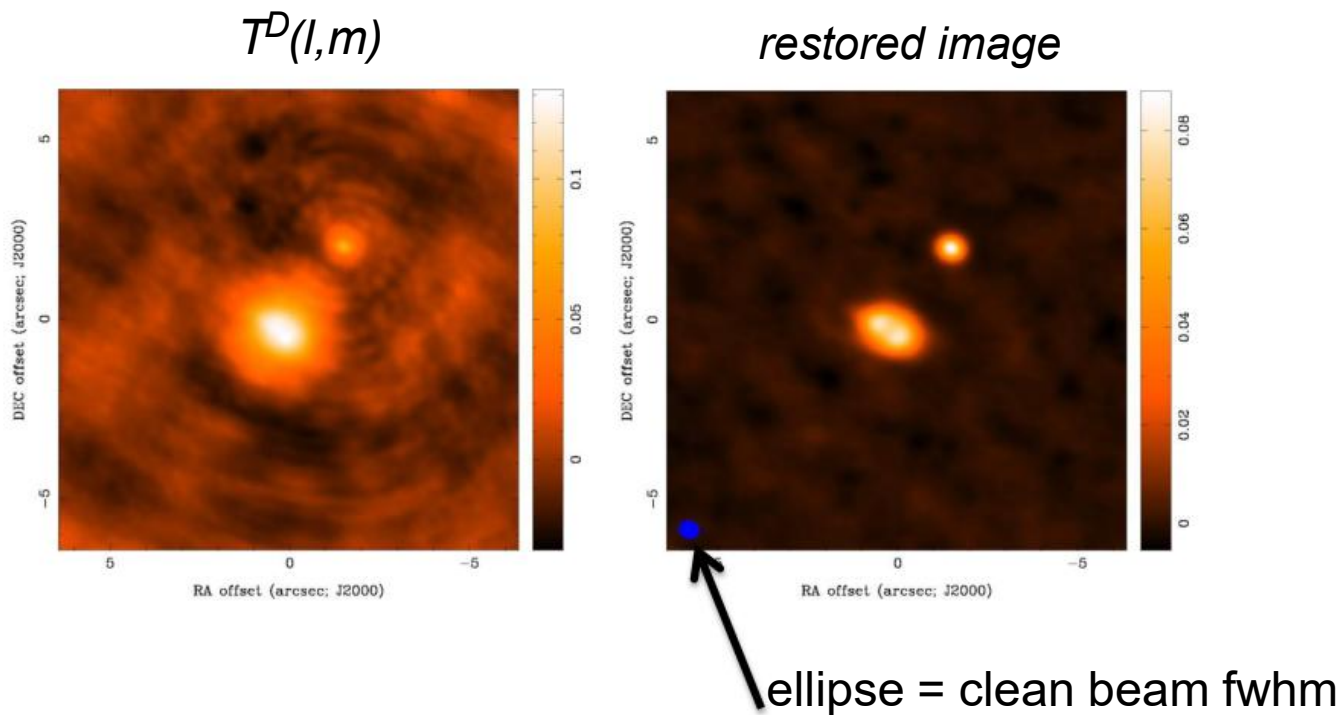
# clean Example

 $T^D(l, m)$ *300 Clean Components**residual map*

# clean Example

 $T^D(l,m)$ *583 Clean Components**residual map*

# clean Example



*final image depends on*

*imaging parameters (pixel size, visibility weighting scheme, gridding)  
and deconvolution (algorithm, iterations, masks, stopping criteria)*



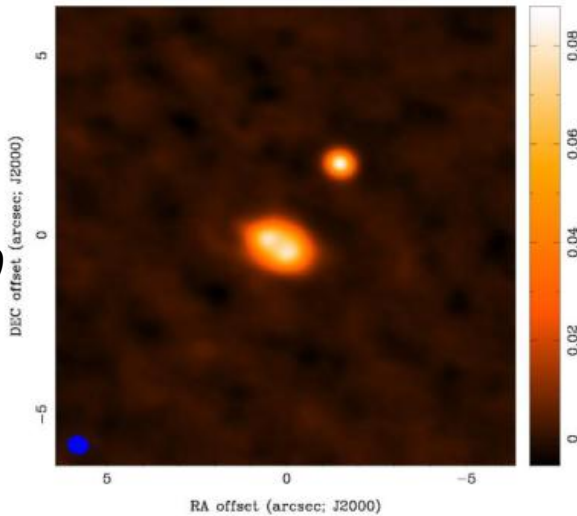
# CASA clean filename extensions

- *<imagename>.image*
  - final clean image (or dirty image if niter=0)
- *<imagename>.psf*
  - point spread function (= dirty beam)
- *<imagename>.model*
  - image of clean components
- *<imagename>.residual*
  - residual after subtracting clean components  
(use to decide whether or not to continue clean)
- *<imagename>.flux*
  - relative sensitivity on the sky
  - pbcor = True divides .image by .flux

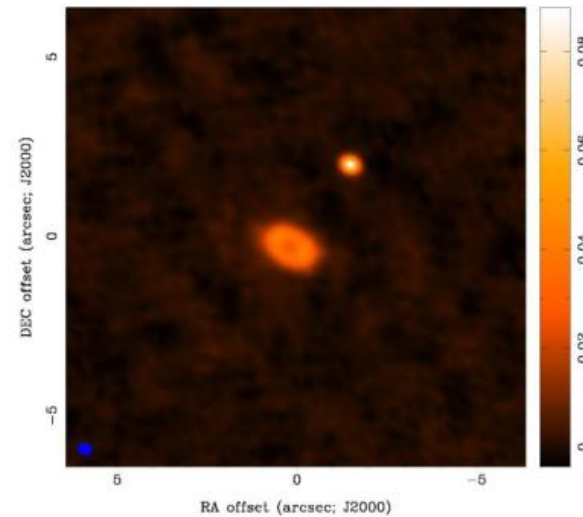


# Results from Different Weighting Schemes

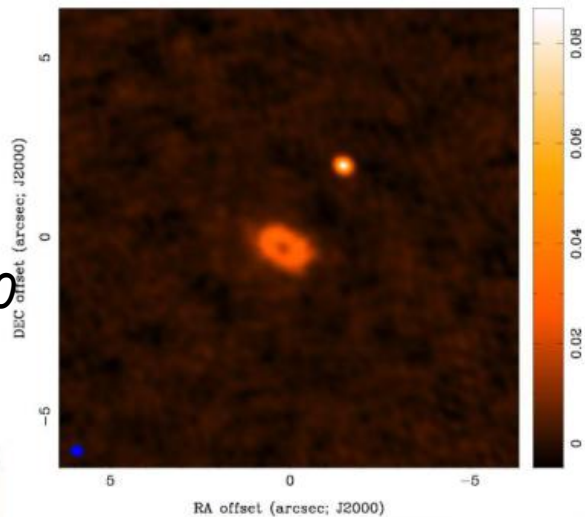
*natural*  
0.59x0.50



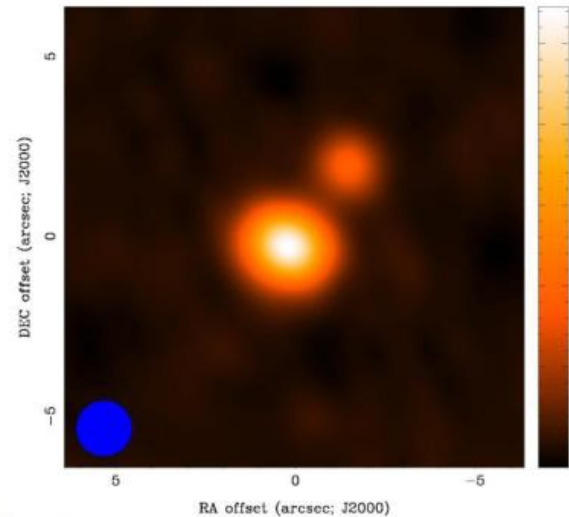
*robust=0*  
0.40x0.34



*uniform*  
0.35x0.30

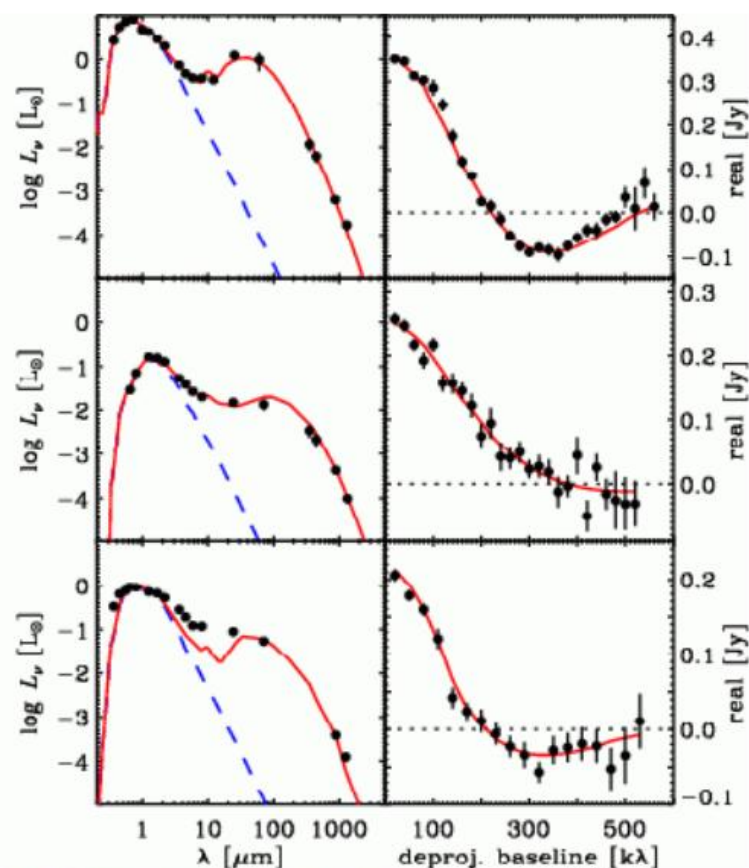
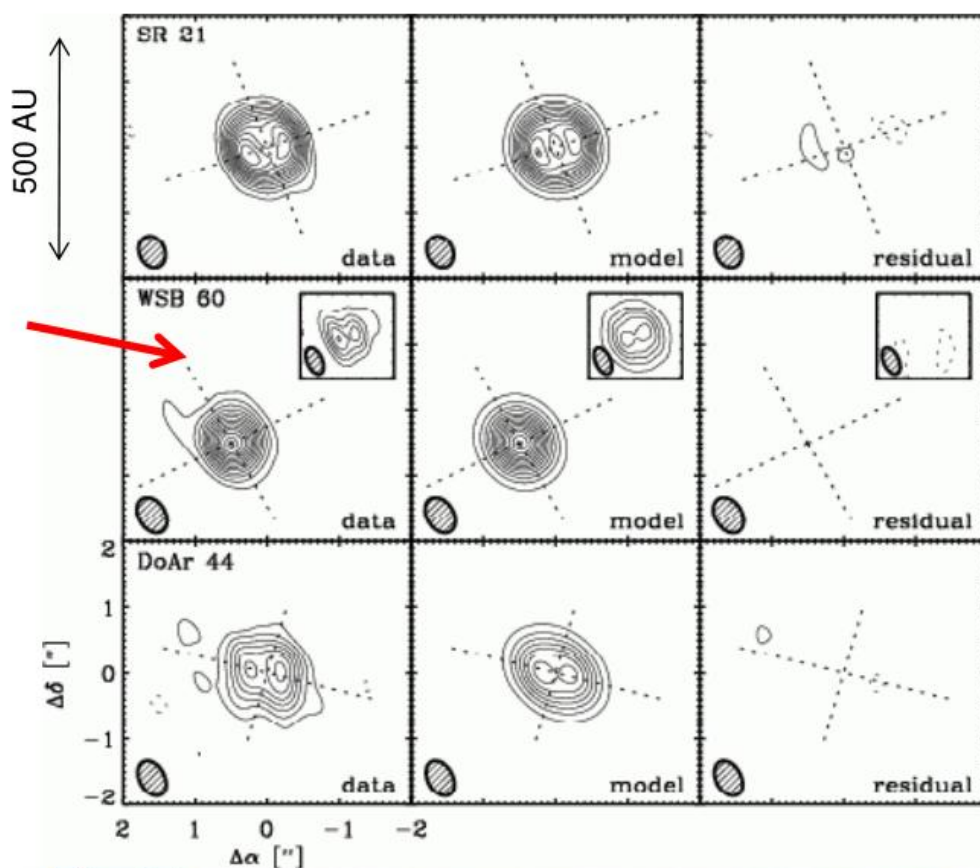


*natural*  
+ taper to  
1.5x1.5



# Tune Resolution/Sensitivity to suit Science

- example: SMA 870  $\mu\text{m}$  images of protoplanetary disk dust continuum emission with resolved inner cavities (Andrews et al. 2009, ApJ, 700, 1502)



# Scale Sensitive Deconvolution Algorithms

- basic clean (or Maximum Entropy) is scale-free and treats each pixel as an independent degree of freedom: no concept of source size
- adjacent pixels in an image are not independent
- an extended source covering 1000 pixels might be characterized by just a few parameters, not 1000 parameters (e.g. an elliptical Gaussian with 6 parameters: x, y, amp, major fwhm, minor fwhm, position angle)
- scale sensitive deconvolution algorithms try to employ fewer degrees of freedom to model plausible sky brightness distributions
- MS Clean (Multi-Scale Clean)
- Adaptive Scale Pixel (Asp) Clean
- yields promising results on extended emission



# “Invisible” Large Scale Structure

- missing short spacings can be problematic for large scale structure
- to estimate? simulate observations, or check simple expressions for a Gaussian or uniform disk (appendix of Wilner & Welch 1994, ApJ, 427, 898)

## *Homework Problem*

- Q: By what factor is the central brightness reduced as a function of source size due to missing short spacings for a Gaussian characterized by fwhm  $\theta_{1/2}$ ?
- A: a Gaussian source central brightness is reduced 50% when

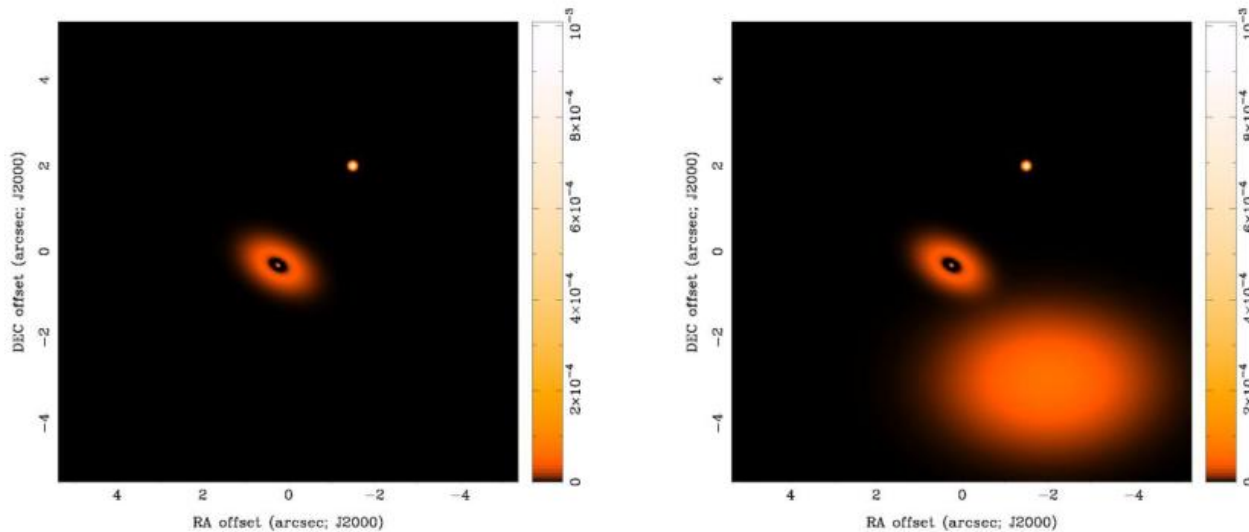
$$\theta_{1/2} = 18'' \left( \frac{\nu}{100 \text{ GHz}} \right)^{-1} \left( \frac{B_{min}}{15 \text{ meters}} \right)^{-1}$$

where  $B_{min}$  is the shortest baseline [meters],  $\nu$  is the frequency [GHz]



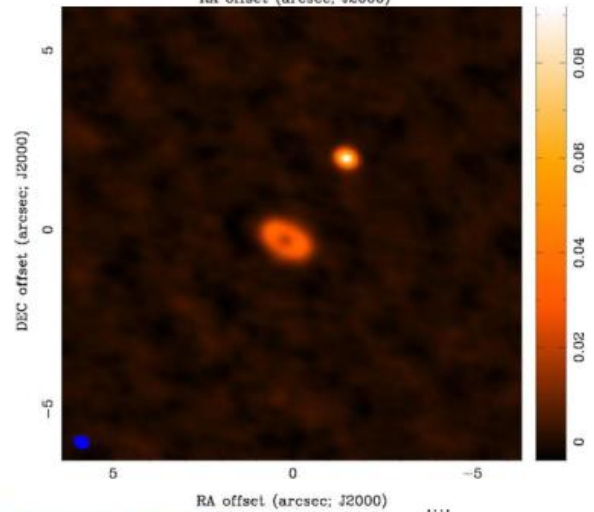
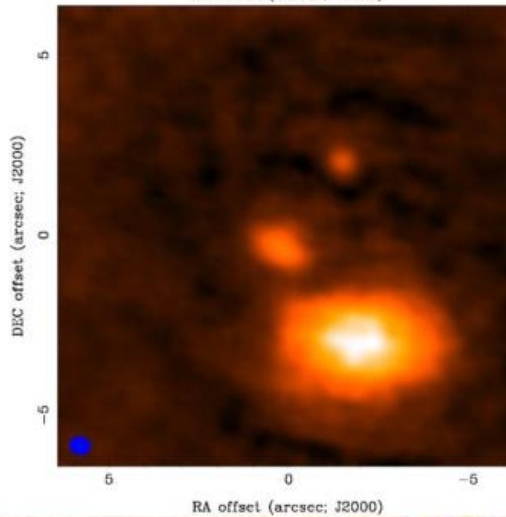
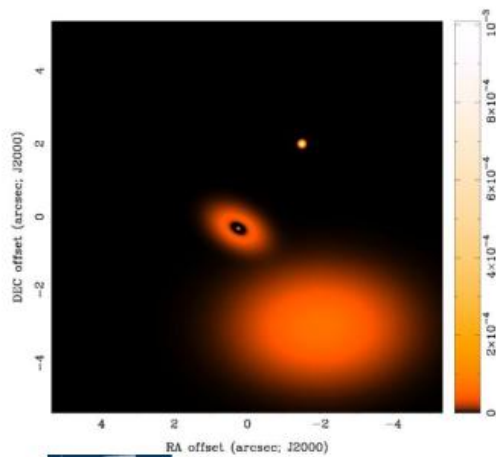
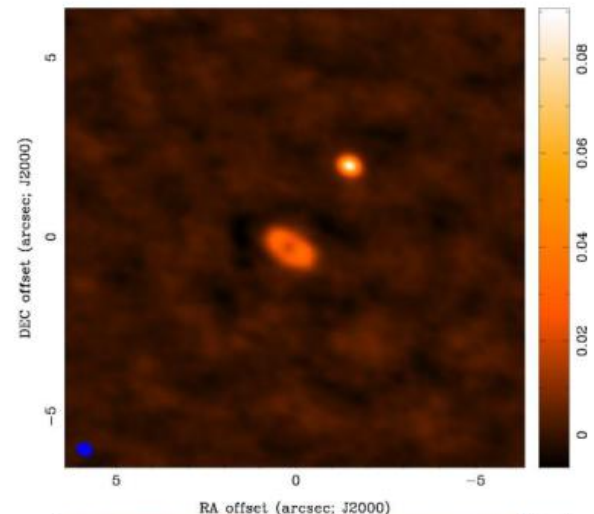
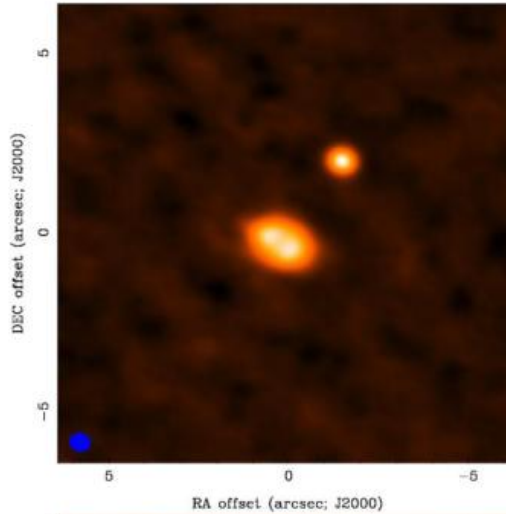
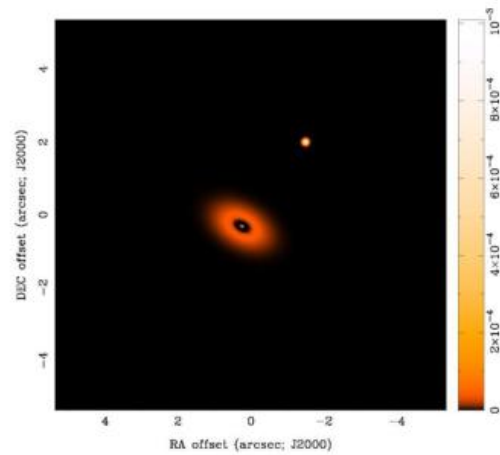
# Missing Short Spacings: Demonstration

- important structure may be missed in central hole of (u,v) coverage
- Do the visibilities observed in our example discriminate between these two models of the sky brightness distribution  $T(l,m)$ ?



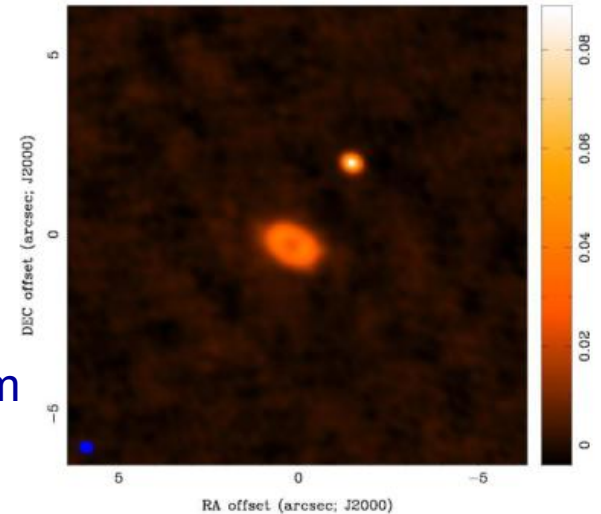
- Yes... but only on baselines shorter than about  $75 \text{ k} \lambda$

# Missing Short Spacings: Demonstration

 $T(l, m)$ 
*natural weight*
 $> 75 \text{ k}\lambda$  *natural weight*


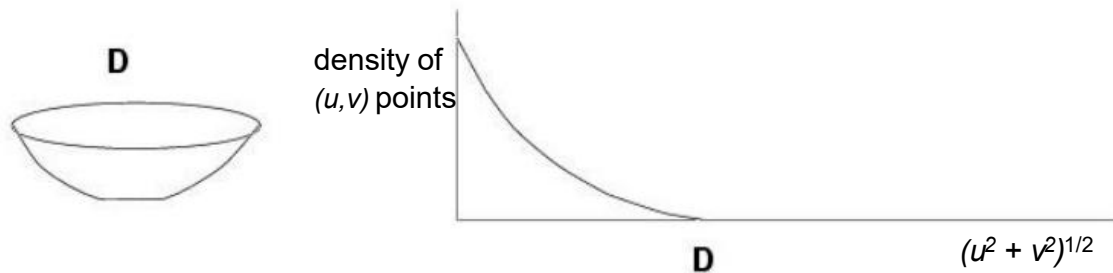
# Measures of Image Quality

- *dynamic range*
  - ratio of peak brightness to rms noise in a region void of emission
  - easy way to calculate a lower limit to the error in brightness in a non-empty region
  - e.g. peak = 89 mJy/beam, rms = 0.9 mJy/beam  
 $\rightarrow DR = 89/0.9 = 99$
- *fidelity*
  - difference between any produced image and the correct image
  - fidelity image = input model / difference
    - =  $\text{model} * \text{beam} / \text{abs}(\text{model} * \text{beam} - \text{reconstruction})$
    - = inverse of the relative error
  - need knowledge of the correct image to calculate



# Techniques to Obtain Short Spacings

use a large single dish telescope



- all Fourier components from 0 to D sampled, where D is dish diameter (weighting depends on illumination)
- scan single dish across sky to make an image  $T(l,m) * A(l,m)$   
where  $A(l,m)$  is the single dish response pattern
- Fourier transform single dish image,  $T(l,m) * A(l,m)$ , to get  $V(u,v)a(u,v)$   
and then divide by  $a(u,v)$  to estimate  $V(u,v)$  for baselines  $< D$
- choose D large enough to overlap interferometer samples of  $V(u,v)$   
and avoid using data where  $a(u,v)$  becomes small, e.g. VLA & GBT



# Techniques to Obtain Short Spacings

use a separate array of smaller antennas

- small antennas can observe short baselines inaccessible to larger ones
- the larger antennas can be used as single dish telescopes to make images with Fourier components not accessible to the smaller antennas
- example: ALMA main array + ACA

main array

50 x 12m: 12m to 14+ km

ACA

12 x 7m: covers 7-12m

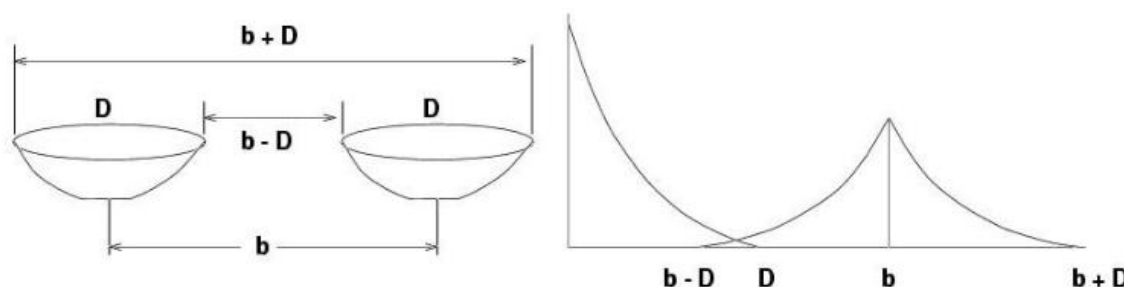
4 x 12m single dishes: 0-7m



# Techniques to Obtain Short Spacings

mosaic with a homogeneous array

- recover a range of spatial frequencies around the nominal baseline  $b$  using knowledge of  $A(l, m)$ , shortest spacings from single dishes (Ekers & Rots 1979)



- $V(u, v)$  is a linear combination of baselines from  $b-D$  to  $b+D$
- depends on pointing direction  $(l_0, m_0)$  as well as on  $(u, v)$

$$V(u, v; l_0, m_0) = \int \int T(l, m) A(l - l_0, m - m_0) e^{i2\pi(ul + vm)} dl dm$$

- Fourier transform with respect to pointing direction  $(l_0, m_0)$

$$V(u - u_0, v - v_0) = \left( \int \int V(u, v; l_0, m_0) e^{i2\pi(u_0 l_0 + v_0 m_0)} dl_0 dm_0 \right) / a(u_0, v_0)$$

# Self Calibration

- *a priori* calibration using external calibrators is not perfect
  - interpolated from different time, different sky direction from source
- basic idea of self calibration is to correct for antenna based phase and amplitude errors *together* with imaging to create a source model
- works because
  - at each time, measure  $N$  complex gains and  $N(N-1)/2$  visibilities
  - source structure can be represented by a small number of parameters
  - a highly overconstrained problem if  $N$  large and source simple
- in practice, an iterative, non-linear relaxation process
  - assume source model  $\rightarrow$  solve for time dependent gains  $\rightarrow$  form new source model from corrected data using e.g. clean  $\rightarrow$  solve for new gains
  - requires sufficient signal-to-noise at each solution interval
- loses absolute phase from calibrators and therefore position information
- dangerous with small  $N$  arrays, complex sources, marginal signal-to-noise



# Concluding Remarks

- interferometry samples Fourier components of sky brightness
- make an image by Fourier transforming sampled visibilities
- deconvolution attempts to correct for incomplete sampling
- remember
  - there are an infinite number of images compatible with the visibilities
  - missing (or corrupted) visibilities affect the entire image
- astronomers must use judgement in the imaging and deconvolution process
- it's fun and worth the trouble → high angular resolution images!

*many, many issues not covered in this talk: see References and upcoming talks*



END

